Synthesis of Peptidoglycan Fragments to Probe the Chemical Biology of the Innate Immune System

by

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ABSTRACT

The bacterial cell wall, comprised of peptidoglycan (PG), is a key tool for the identification of bacteria by the innate immune system. For the body to function properly, this system needs to be able to accurately dispose of pathogenic threats while retaining commensal bacteria. To accomplish this, immune receptors can recognize PG from a variety of bacterial species and differentiate between them. Unfortunately, errors are known to occur in this recognition, leading to false inflammatory responses. For many people, this occurs in the gut, resulting in inflammatory bowel diseases (IBD's). The immune proteins NOD1 and NOD2 (nucleotide oligomerization binding proteins) have been found to bind PG fragments known as muramyl dipeptide (MDP), and iE-DAP respectively. Both are PG derived ligands which have been studied frequently. However, when bacteria are digested in the body, PG can fragment into a diverse pool of structures. It's important to have an expansive set of chemical tools which allow for the investigation of innate immune response with various PG fragments. Here, synthesis was focused on obtaining a library of muramyl dimers linked through their peptide chain, a class of compounds whose properties have been scarcely studied. The synthesis of lysine crosslinked dimers was previously displayed in our lab using solid phase techniques. This work expands on the aforementioned synthesis by implementing *meso*-diaminopimelic acid (m-DAP) crosslinking patterns which are characteristic to Gram-negative bacteria. Furthermore, a solution phase approach has been devised. The completed synthesis of these compounds will open the doors to an array of applications for muramyl dimers, one such being their use in qRT-PCR assays. Furthermore, synthetic efforts were also directed toward the creation of an alkyne linked MDP derivative. Through previously developed methyl N,O,-

hydroxylamine linker chemistries, successful synthesis of this compound will allow for rapid functionalization of MDP.

Chapter 1

INTRODUCTION

1.1 The Innate Immune System and Human Microbiome

The human innate immune system is a complex machine with a litany of tasks. One such task is the mediation of host interactions with the microbiome. The microbiome consists of up to 100 trillion microbial cells with an estimated 38 trillion[1] bacteria serving as one of the primary constituents[2]. On a daily basis, immune cells sort through a countless number of these bacteria, determining which are commensal and which are pathogenic. Commensal strains of bacteria can serve various mutualistic roles for the body in areas such as metabolism and even pathogen defense[3,4]. The opposite is true for pathogenic bacteria where their colonization can sometimes perpetuate serious disease. As previously stated, the innate immune system is one of our primary tools for differentiating between these general types of bacteria. However, this system is error prone in some individuals, particularly within the gut microbiome. Host genetic mutations are thought to permit the misrecognition of commensal bacteria, leading to false inflammatory responses and disorders such as Crohn's disease[5]. In hopes of developing therapies against such disease, the chemical and biological interactions between microbiome and immune system must be further elucidated.

1.2 Peptidoglycan Structure and Significance

The anatomy of a bacterial cell is different than that of a mammalian cell on many fronts, one of the most important being the presence of a carbohydrate-based cell wall. This protective coat provides bacteria with much needed defense from physical harm, as well as proper rigidity to maintain pressure within the cell[6]. However, innate immune receptors have evolved to use this defining feature to their advantage, utilizing structural characteristics of the cell wall to identify bacteria. One of the most important components of the cell wall is the polymer known as peptidoglycan (PG), pieces of which are major ligands for immune receptors.

There are two general classes of bacteria: Gram-positive and Gram-negative. One of the major differences between the two is the format of their PG protection. Gram-positive species feature a cell membrane surrounded by a thick layer of PG

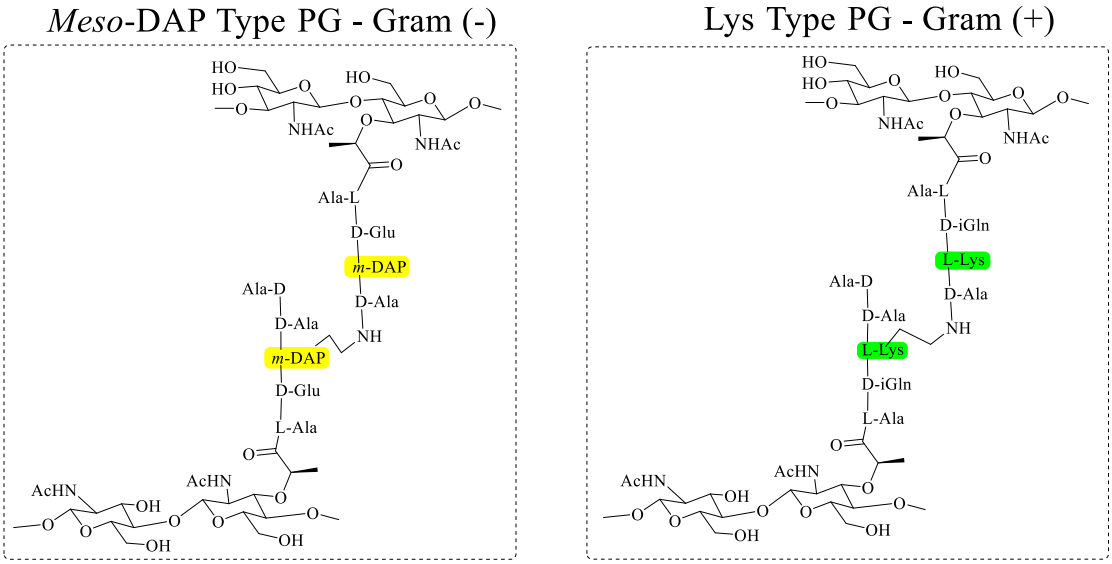


Figure 1.1 Generic chemical structure of Gram-positive and Gram-negative peptidoglycan. Lysine is highlighted in Gram-positive whereas *m*-DAP is highlighted for Gram-negative.

whereas Gram-negative possess a membrane, followed by a thin layer of PG and then another outer membrane. Despite these differences, the overall chemical structure of PG remains largely conserved. In most instances, it consists of alternating units of β-(1-4) linked N-acetylmuramic acid (NAM) and N-acetylglucosamine (NAG) which are crosslinked by a peptide chain (Figure 1.1). The peptide chain usually features an L-alanine residue in the first position connected to the lactoyl moiety of NAM. In the second position is D-isoglutamine followed by either L-lysine or *meso*-diaminopimelic acid (*m*-DAP) in the third. Lysine tends to be characteristic of Gram-positive bacteria whereas *m*-DAP is found more often in Gram-negative, a property which is crucial to recognition[7]. From here, the peptides are crosslinked to complete generic PG scaffold.

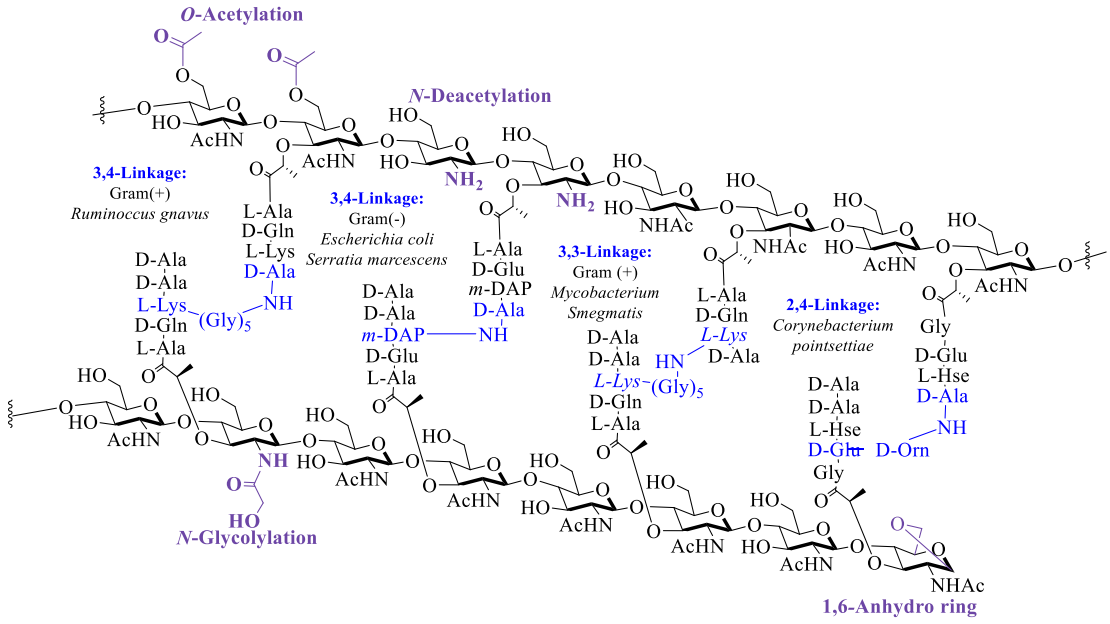


Figure 1.2 Linkages and modifications observed in the peptidoglycan structure of various bacterial species[8].

Although this structure is well conserved, there are a plethora of PG alterations found in nature which help bacteria evade host defenses. For example, both NAG and NAM can be decorated with different modifications such as N-glycolylation[9] or O-acetylation[10]. Similarly, the peptide component can also undergo alteration, most often seen in the class of crosslinking. 3-4 linkages, indicating a bond between residues in the third and fourth position of their respective sides, are among the most frequently observed. Other varieties such as 3-3 and 2-4 linkages are also found in select species[8]. When considering the interactions between bacterial peptidoglycan and the host immune system, these modifications become significant to recognition.

1.3 Recognition of Peptidoglycan by the Immune System

As established previously, potential pathogenic bacteria are constantly interrogated by immune defenses in the gut microbiome. Pattern recognition receptors (PRR's) mediate this response by interacting with anatomical structures possessed solely by bacteria[11]. These features are known as pathogen associated molecular patterns (PAMP's) and PG falls into this category.

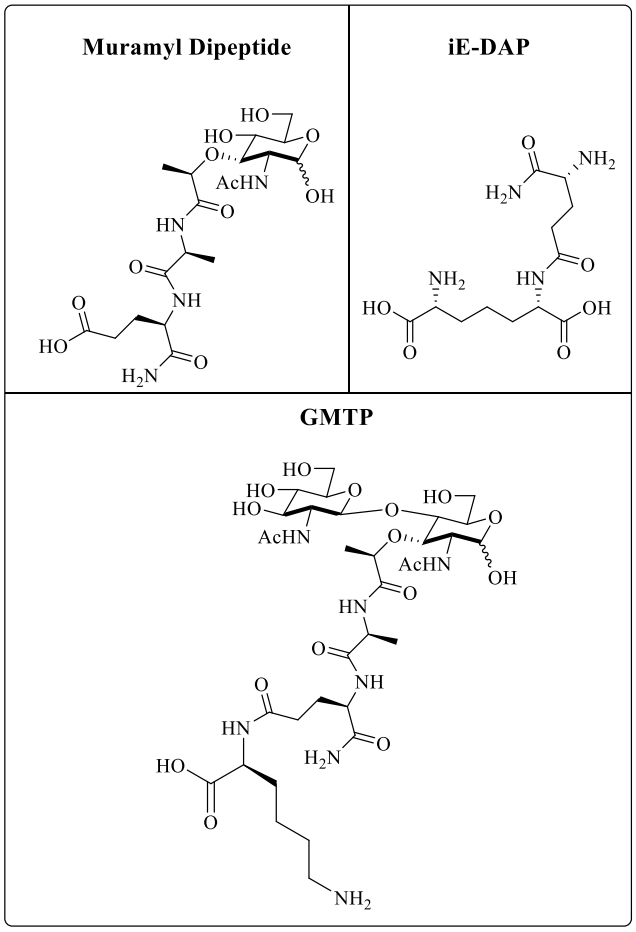


Figure 1.3 Chemical structures of muramyl dipeptide (Detected by Nod2), iE-DAP (Detected by Nod1), and the disaccharide fragment GMTP.

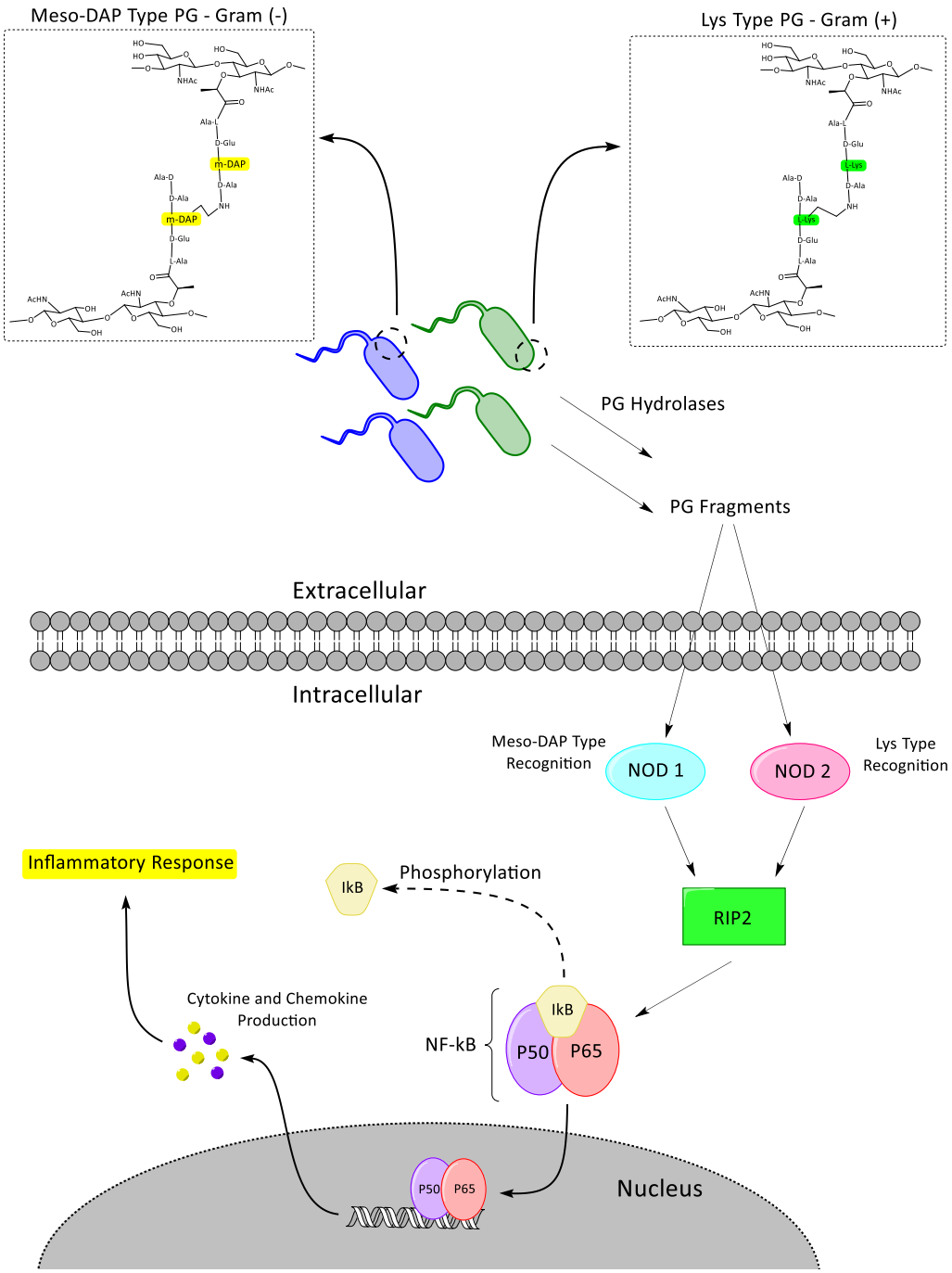


Figure 1.4 Nod1 and Nod2 activation of of NF-κB with the following inflammatory response[12].

As bacteria grow, or are digested by host enzymes (i.e., lysozyme), pieces of the cell wall are shed, resulting in fragments of varying chemical structure. PRR's have evolved to recognize specific cell wall pieces, leading to a downstream inflammatory response. Both Toll-like receptors (TLR's) and NOD-like receptors (NLR's) belong to the PRR family. TLR's are membrane bound while NLR's are cytosolic, however, both work in concert to respond to virulent bacterial threats. In recent years, the specific ligands associated with NLR's Nod1 and Nod2 have been elucidated. Nod1 is generally responsible for the recognition of PG fragments containing *m*-DAP, pertaining mostly to Gram-negative species. More specifically, γ-D-glutamyl-meso-diaminopimelic acid (iE-DAP) (Figure 1.3) is the crucial piece for Nod1 detection[7]. Nod2 serves a similar purpose where it's been found to directly bind the fragment known as muramyl dipeptide (MDP)(Figure 1.3)[13]. In both Nod1 and Nod2, these detection events trigger an inflammatory response via the NF-κB pathway[14,15]. Due to commercial availability, much work has been done to elucidate the interactions of MDP and iE-DAP with their respective receptors. However, but there are far more PG derived ligands which are suspected to illicit immune response[16]. This includes fragments featuring atypical amino acids, disaccharides, and crosslinked dimers. Compounds of this nature are difficult to obtain and synthetically challenging, leading to a lack of investigation.

Recent literature has shown the value of expanding the PG toolbox to deepen our understanding of immune recognition. Grimes, Reinecker and coworkers established that the disaccharide fragment known as N-acetylglucosamine N-

acetylmuramic acid dipeptide (GMTP) (Figure 1.3) was prominent in the cell lysate of the bacterium *Lactobacillus acidophilus*. This led to the development of synthetic methodology to obtain GMTP for in depth characterization, and gene transcriptional analysis. Using a quantitative reverse transcriptase PCR (qRT-PCR) assay, it was revealed that GMTP was a more potent immune stimulator when compared to MDP[16]. Further, this led to the discovery of a new inflammatory bowel disease (IBD) associated gene cluster, indicating the clear importance of next generation synthetic PG adjuvants.

1.4 Specific Aims

Here, the synthesis and significance of crosslinked muramyl dimers was investigated. Compounds of this nature feature dimerized carbohydrates linked through varying peptide cross bridges, looking to mimic structures found in nature. A workflow was adopted where certain strains of bacterium, such as *Neisseria meningitidis*[17], have been shown to release dimer PG fragments. As previously stated, the immunogenicity of such compounds is scarcely studied, making this a worthwhile investigation. From here, the dimer fragments are to be analyzed for their ability stimulate inflammatory gene clusters using qRT-PCR. Additionally, the synthesis of functionalized MDP derivatives was investigated. As previously stated, MDP is an essential tool for PG analysis, but derivatives remain difficult to obtain. A simple synthetic method featuring a one-step neoglycosylation was explored to append an alkyne handle to MDP off the C-1 position.

Chapter 2

SYNTHESIS OF *m*-DAP CROSSLINKED MURAMYL DIMERS

2.1 Dimer Background

2.1.1 Dimers in Literature

Peptidoglycan dimer compounds are characterized by sugar moieties linked through varying peptide chains. Based on the structure of peptidoglycan, it is possible for fragments of this nature to be released as the β-(1-4) linkage is cleaved.

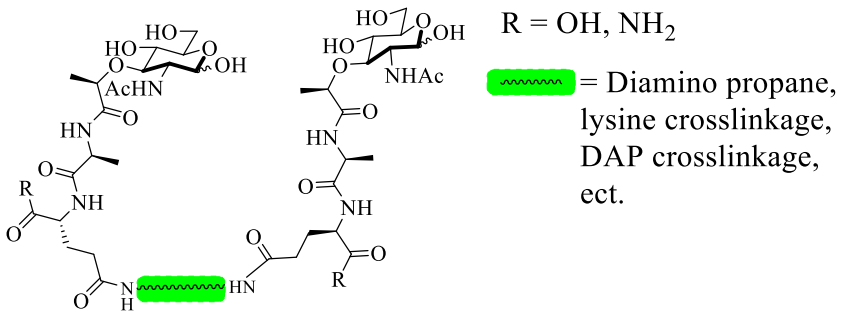


Figure 2.1 Generic structure of a muramyl dimer compound.

Work from Murata and colleagues in 1996 supports the concept of immune stimulation from dimer compounds. Their approach investigated GADP dimers, GADP being the glucose analog of MDP. At this point, it had already been determined by Lederer and coworkers that MDP was the minimal synthetic PG adjuvant needed to elicit immune response[18]. Additionally, it had also been found that GADP was a more potent stimulator, leading to the synthesis of GADP dimers as opposed to MDP[19].

Through analyzing glucose consumption of macrophage like cells, it was determined that GADP dimers provided even greater immune stimulation than monomeric GADP, suggesting the idea of a multivalent receptor[20]. At this time, Nod2 was not yet known to be the receptor for MDP, making it quite possible that it also bound the dimers analyzed here. This concept was put to the test in work from Fukase and coworkers. They designed an elegant synthesis to obtain lysine crosslinked dimers as well as disaccharide dimer PG fragments. Interestingly, the disaccharide dimer compounds they obtained were not shown to be potent activators of Nod2 via NF-κB luciferase reporter assay[21]. Furthermore, data was not presented regarding the activation profile of monosaccharide dimer fragments. As seen in the work from Grimes and Reineker, gene transcriptional analysis with qRT-PCR provides a valuable tool for assessing the role of a PG fragment, indicating it would be useful to subject dimer fragments to this assay.

2.1.2 Dimers in the Grimes Lab

In the Grimes lab, the synthesis of dimers was first investigated by former group member Dr. Lauren Genova. Fukase and colleagues provided compelling evidence to support that dimers are not major ligands of Nod2, however, questions remained. Their synthesis featured a propyl moiety off the anomeric carbon, a group not native to PG structure. Chemically, propyl groups are drastically different than the standard hydroxyl making this a significant difference. Additionally, they presented dimers with $_{D}$-isoglutamine at the second position while $_{D}$-glutamate is also quite

prominent. These changes merited the further investigation of dimers. Along with this, MDP was only shown to directly bind with Nod2 in 2012, after the work from Fukase. Subjecting dimers to the SPR assay developed by the Grimes group[13] would provide definitive proof if a binding interaction exists. This evolved into a library of dimers to be synthesized featuring diamino and pentaglycine linkers based on molecular models[22]. The approach displaying a simple diamino linker was continued to completion by former group members Thomas Harmon, and Dr. Klare Bersch. Utilizing chemistries previously developed by the Grimes group, they were able to show the synthesis of what was termed a "2,2" linked dimer, indicating a linkage between D-isoglutamine residues, both in the second position.

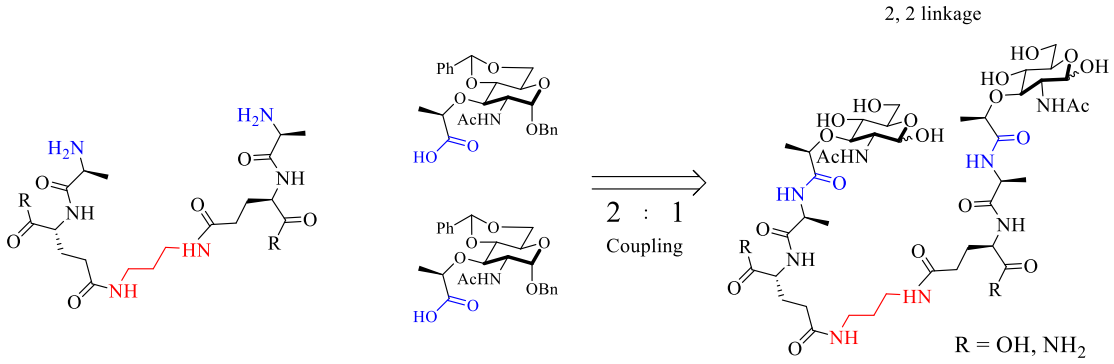


Figure 2.2 Retrosynthetic scheme for solution phase synthesis of a "2,2" linked dimer. Highlighted in red is the 1,3 diamino crosslinker, highlighted in blue are the amine and carboxylic acid groups which participate in the 2:1 coupling step.

Multiple synthetic routes were designed, hinging around a 2:1 coupling step between protected MDP, and the 1,3 diamino propane linker. This compound was useful in

showing the success of a solution phase synthesis but failed to replicate structures which might be observed in nature.

2.1.3 Lysine Linked Dimers

To address the shortcomings of the 2,2 linked dimer, we proposed the synthesis of a next generation library featuring more complex linkages which better mimic native PG structure. The first variety incorporated lysine cross bridges. Based on PG structure, these compounds looked to replicate dimers released from Gram-positive bacterium. This was initially approached from a solution phase perspective, similar to what was designed when synthesizing the 2,2 dimer. The “di-lysine” bridge and protected of $_{L}$-Ala-$_{D}$-Glu dipeptide were synthesized separately and then joined a 2:1 coupling step.

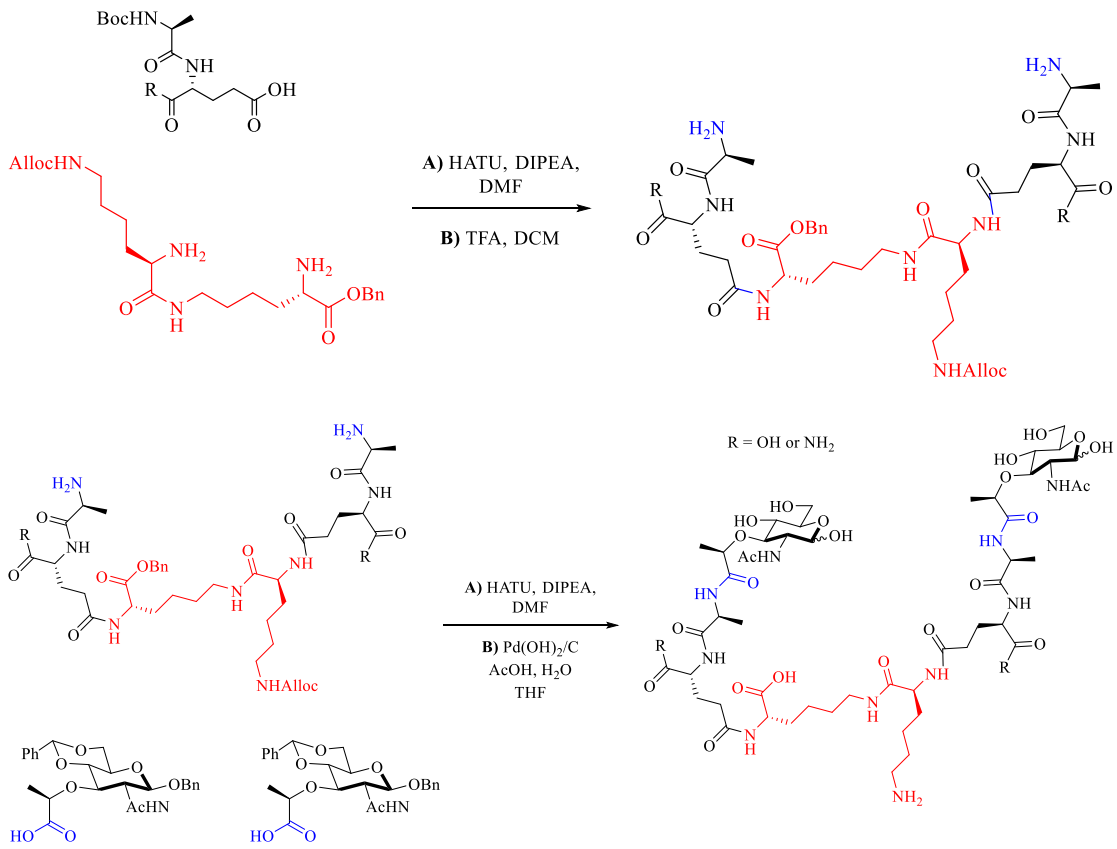
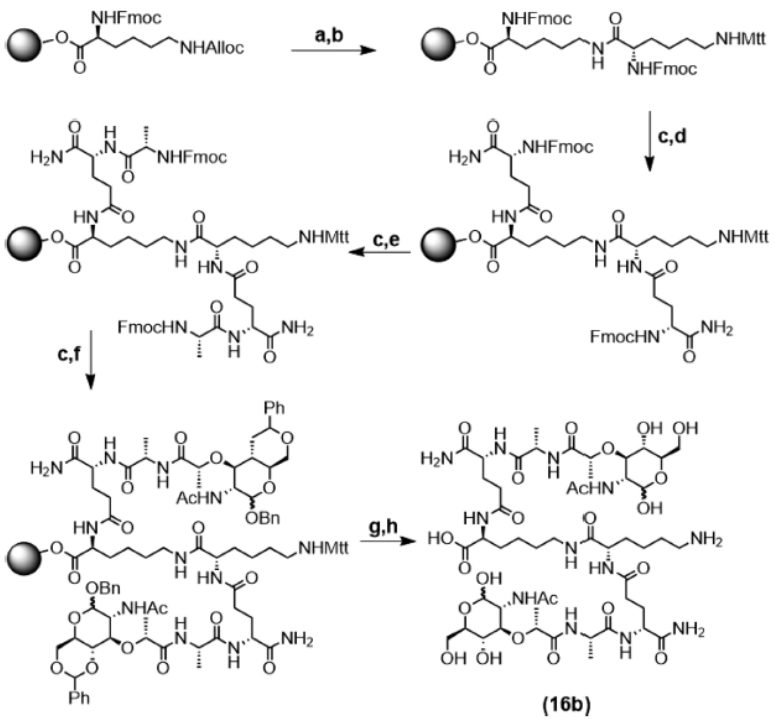


Figure 2.3 Low yield solution phase synthesis schematic for a 3,3 lysine crosslinked dimer. In red is the lysine crosslinker, in blue are the amine and carboxylic acid groups involved in 2:1 coupling.

From there, protected NAM was installed to yield the full protected dimer. Mashayekh, Bersch and I contributed on this front, but major solubility issues were encountered once the full scaffold was complete. The product was not soluble in common organic solvent, leading to very poor yield in the subsequent deprotection steps[23]. To overcome this, a solid phase approach was spearheaded by Mashayekh. In this approach, L-Lysine was coupled to 2-chlorotrityl resin. Subsequent deprotections and coupling yielded either the 3,3 or 3,4 linkage depending on the presence of D-Ala.



[a]Conditions: (a) $Pd(PPh_3)_4$, $PhSiH_3$, DMF; (b) Fmoc-L-Lys(Mtt)-OH, HATU, DIPEA, DMF; (c) 20% piperidine in DMF; (d) Fmoc-D-Glu-NH_2, HATU, DIPEA, DMF; (e) Fmoc-L-Ala-OH, HATU, DIPEA, DMF; (f) **3**, HATU, DIPEA, DMF; (g) TFA/H_2O/TIPS (95:2.5:2.5); (h) Pd/C, H_2O, AcOH, H_2, 87%.

Figure 2.4 Solid phase dimer synthesis designed and executed by Mashayekh[23].

From there, the remainder of the synthesis relied on symmetrical building of the peptide on each side, and then capping each chain with NAM. This approach was successful, delivering a synthetic route for both 3,3 and 3,4 lysine crosslinked dimers[23]. Furthermore, derivatives swapping D-isoglutamine for D-glutamate were also synthesized. Preliminary results have yielded interesting data, including significant NF-κB activation from the D-glutamate dimers[24].

2.1.4 Proposal

With the success of lysine linked dimers, we determined that the synthesis of *m*-DAP linked dimers was a necessary next step. As previously stated, lysine cross bridges looked to simulate fragmentation patterns from Gram-positive species. DAP linkages would offer the opposite, interrogating gene activation when stimulated with a Gram-negative mimic. In the instance of the lysine dimer synthesis, the work of Fukase served as a precedent. Conversely, there is no literature reported synthesis which incorporates *m*-DAP into dimer PG fragments, increasing the significance. To overcome this, we investigated the synthesis of a 3,3 *m*-DAP linked muramyl dimer.

2.2 Synthesis

In the instance of previously published dimers, obtaining the lysine crossbridge was not a significant challenge. Lysine with appropriate protecting groups can be purchased commercially, greatly decreasing the difficulty of the synthesis. This is not the case with respect to *m*-DAP, a unique amino acid which cannot be obtained easily. Furthermore, incorporation into dimers requires an orthogonal protecting group strategy, further complicating the synthesis. Fortunately, this was overcome by Geert Jan Boons and colleagues in 2005. In an elegant synthesis, they crafted the DAP scaffold through a Grubbs cross metathesis reaction between allyl and vinyl glycine derivatives[25]. This approach is highly advantageous as it offers stereocenter control, along with the ability to easily install protecting groups.

In approaching the synthesis of *m*-DAP linked dimers, we decided to again investigate a solution phase synthesis. Despite the lack of success using solution phase with the lysine dimers, we hypothesized that the addition of lipophilic protecting groups would help alleviate issues with solubility.

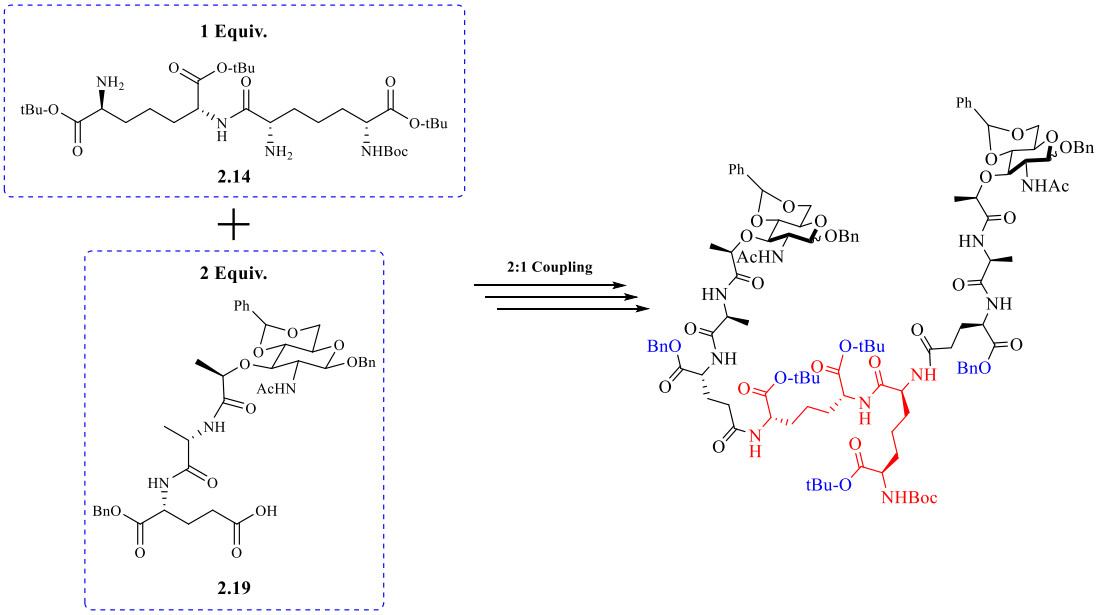


Figure 2.5 2:1 coupling envisioned for solution phase 3,3 *m*-DAP dimer. Highlighted in red is the "di-DAP" crosslinker, in blue are the additional hydrophobic protecting groups to promote solubility.

These groups are featured on the extra carboxylic acids of DAP (as compared to lysine), as well as on the D-glutamic acid residues from MDP.

2.2.1 "Di-DAP" Synthesis

From here, the synthesis was divided into two portions: protected MDP and the "di-DAP" crosslinker. Work started on the DAP fragment where an "east to west"

approach was devised by Siavash Mashayekh. As shown in figure 2.8, the structure grows from the east to the west, resulting in two *m*-DAP pieces linked together as they might be in native PG. Allyl glycine was obtained commercially and then outfitted with adequate protecting groups. Vinyl glycine, however, must be synthesized from aspartic acid starting materials, something also shown by the Boons group. Reduction of the γ carboxylic acid followed by dehydration of the alcohol results in a vinylic alkene[25].

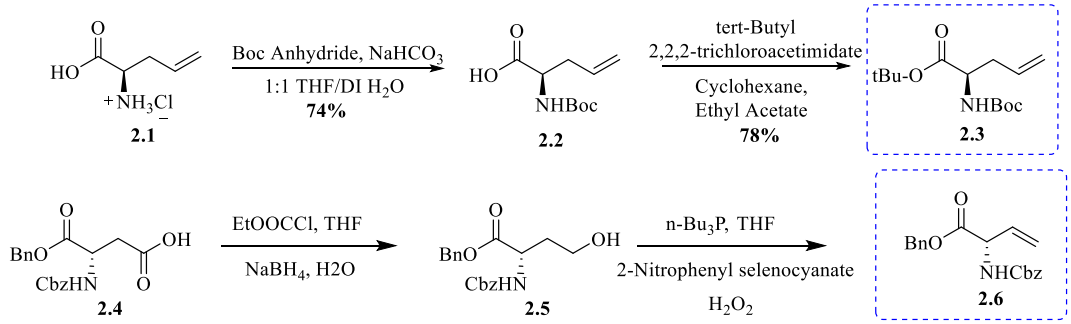


Figure 2.6 Synthesis of vinyl and allyl glycine components.

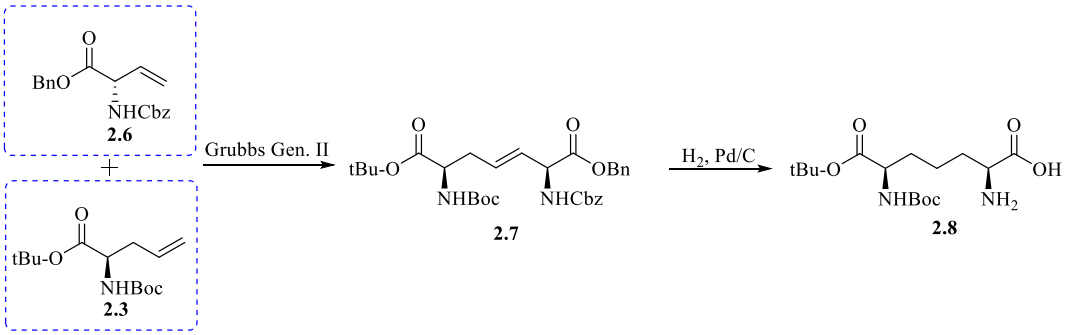


Figure 2.7 Grubb's cross metathesis and subsequent hydrogenation between allyl and vinyl glycine derivatives.

These two structures were then stitched together using Grubb's second-generation catalyst, providing the scaffold for *m*-DAP. Following this, standard palladium catalyzed hydrogenation removed both the alkene, benzyl carbamate (Cbz), and benzyl ester (Bn). In order to continue the east to west synthesis, the amine had to be reprotected with Cbz, leaving just the left most carboxylic acid available for coupling.

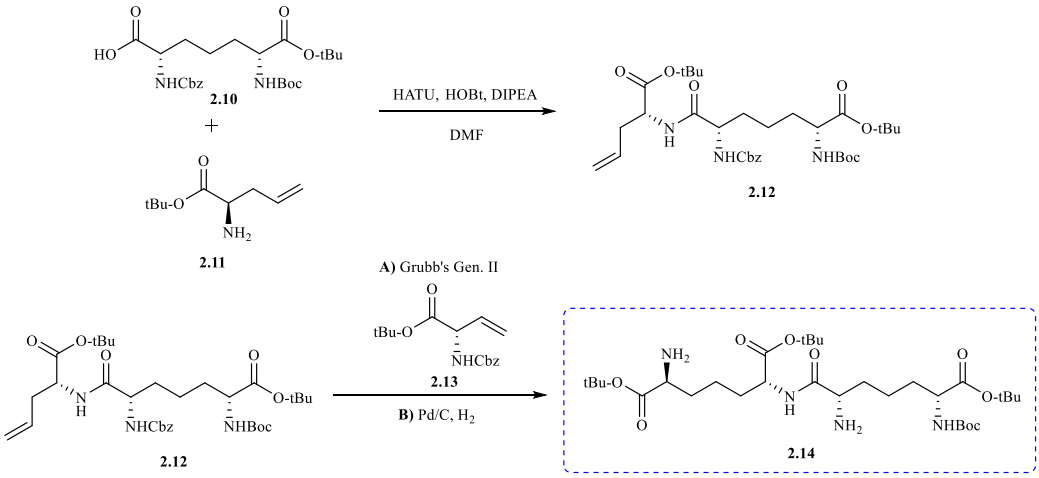


Figure 2.8 "East to west" synthetic approach designed by Mashayekh. The "di-DAP" crosslinker grows from the right to the left through subsequent coupling and cross metathesis reactions.

This allowed for the addition of another allyl glycine unit, followed by a second Grubbs reaction. Currently, this reaction is in the process of being purified and once it is complete, the full "di-DAP" scaffold will be obtained. With the backbone completed, hydrogenation will remove the alkene as well as both Cbz groups needed for the penultimate 2:1 coupling to MDP.

2.2.2 Protected MDP Synthesis

MDP was synthesized in a manner similar to what had been previously reported[26,27]. Protected NAM was obtained using well established methodology from the Grimes group[26,27]. The dipeptide component started with commercially available α benzylated $_{D}$-glutamic acid. This was coupled to NHS activated boc-$_{L}$-alanine, followed by TMS ethanol protection of the free λ carboxylic acid. Subsequently, boc

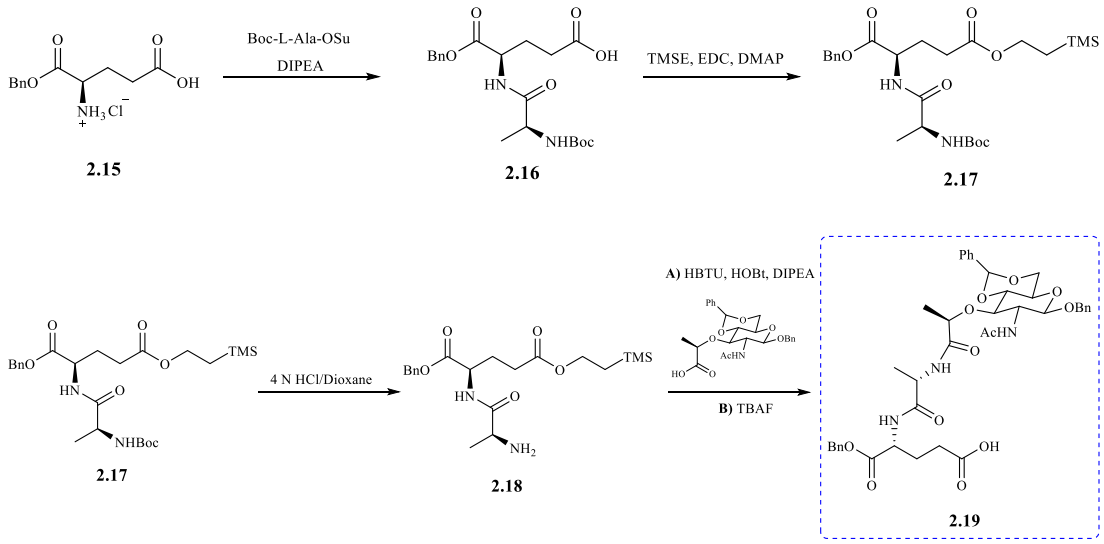


Figure 2.9 Synthetic route to obtain protected MDP.

was removed from alanine to open a free amine for coupling to protected NAM. Interestingly, we found that reactions utilizing TFA were not compatible with the TMS group installed on glutamic acid. In the presence of TFA, both the removal of boc and cleavage of the TMS ester was observed. This resulted in a switch to 4 N HCl in dioxane as the reagent for boc removal. After troubleshooting, remaining steps were NAM coupling and TMS deprotection, both of which proceeded as expected. This

resulted in a fully protected MDP derivative with a carboxylic acid available for coupling to the DAP bridge.

With protected MDP in hand and “di-DAP” soon to be obtained, the following step is one of the most important in the synthesis: 2:1 coupling. In this instance, we will opt to use the coupling agent PyOxime. Similar to HATU, it’s known for high efficiency and the minimization of other side reactions. Assuming this reaction proceeds with respectable yield, the final steps will be universal deprotection first with acid, followed by hydrogenation to provide the final product as shown in figure 2.5.

Chapter 3

SYNTHESIS OF A METHYL N,O-HYDROXYLAMINE MDP LINKER

3.1 Linker Background

Probing the interactions of the immune system with the microbiome requires effective chemical tools. As previously explained, MDP is one of the most important and widely studied synthetic PG ligands. Its binding interaction to Nod2 has been well documented through the use of synthetic derivatives such as 6-amino MDP[28]. However, access to modified MDP variants is limited, requiring significant knowledge in organic chemistry to obtain them. This obstacle was a driving force behind the work on methyl N,O-hydroxylamine linkers from Dr. Klare Bersch. Her work displayed both a short three carbon linker, and a long hydrophilic PEG linker which could be appended to C1 of MDP in one step[26].

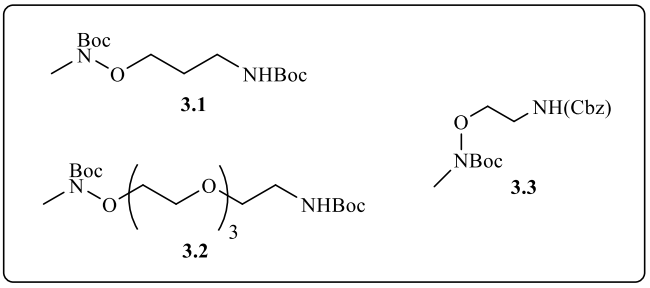


Figure 3.1 Methyl N,O-Hydroxylamine linkers developed to date. Linkers 3.1 and 3.2 were published by Bersch in 2019, linker 3.3 was developed shortly after and is the subject of this work[26].

Originally developed by Peri and colleagues[29], these “neoglycosylation” reactions take place in mild conditions and offer high stereoselectivity toward the β anomer. Furthermore, it was shown that the new C1 linked MDP retained the ability to activate the NF-κB pathway[26]. This led us to develop a third linker compound with two carbons and a different protecting group strategy.

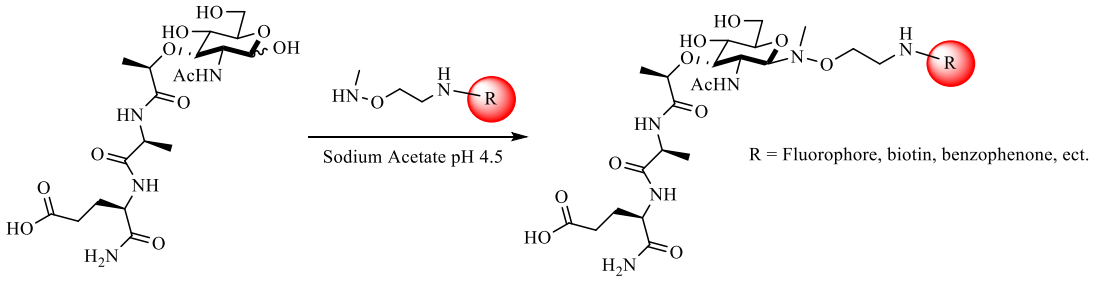


Figure 3.2 “Neoglycosylation” reaction between methyl N,O-hydroxylamine linker and MDP

The benefit provided is functionalization of the linker before appending to MDP, resulting in one less HPLC purification step. For the purpose of wide application, as well as retaining a structure similar to the published linkers, the synthesis of an alkyne handle was investigated in this work.

3.2 Synthesis

The new linker offers a slightly different protecting group strategy, featuring a Cbz group on the primary amine. Using this, the Cbz group can be selectively removed with palladium catalyzed hydrogenation, allowing for its use any desired coupling reaction. From there, the remaining Boc would be removed, freeing the

secondary amine for neoglycosylation to the sugar. Unfortunately, an ethanolamine starting material that is both Cbz protected and substituted with a leaving group is not commercially available. This resulted in the synthesis starting with the bromination of Z-glycinol. The hydroxyl group was first mesylated, and after all the starting material had reacted, an excess of lithium bromide was added. Extra care was taken during this reaction to keep the environment anhydrous. This included flame drying all glassware and needles, and drying solvents with molecular sieves. Any traces of water could

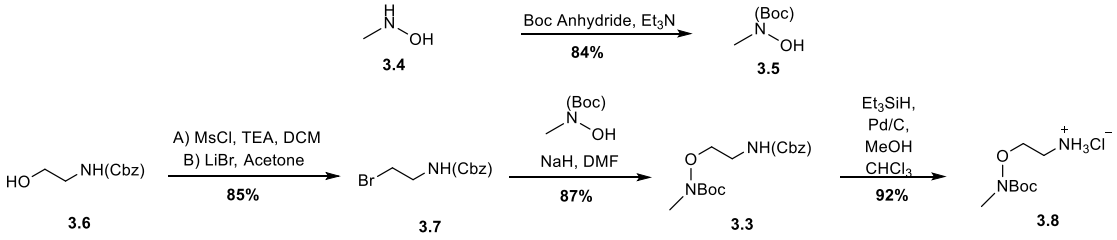
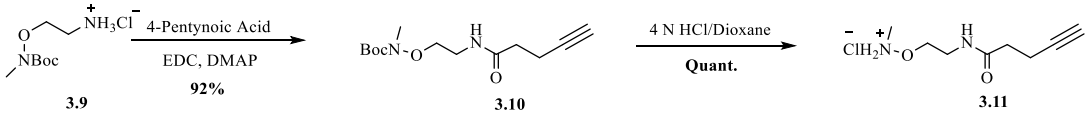


Figure 3.3 Synthetic route to obtain two carbon Cbz linker

have subjected the newly brominated compound to hydroxylation. Once the reaction was completed and purified via silica gel, the methyl N,O hydroxyl moiety could be attached. First however, the N-methylhydroxylamine had to be boc protected. This involved simply reacting boc anhydride with N-methylhydroxylamine and triethylamine. The protected product was now used with the previously shown brominated ethanolamine to produce the fully protected linker. This required generating a nucleophilic alkoxide using sodium hydride and replacing the bromine in an S_N2 style reaction. Again, care was taken to ensure the reaction was anhydrous, especially considering the presence of sodium hydride. Upon completion, excess base

was quenched, and the product was purified via silica gel column chromatography to yield the protected linker. This product is stable over long periods of time when kept at low temperature and can be produced in bulk quantities, allowing it to be stored until needed.

With the linker in hand, the remaining steps pertained to functionalizing with an alkyne handle. Early on, a biotin handle was originally planned as well. However, difficulties were encountered when attempting neoglycosylation with the biotin linker. It was hypothesized that the smaller alkyne probe would be more compatible with that step, as well as providing more general utility as a probe. In a short three step synthesis, I was able to obtain the desired alkyne linker as a chloride amine salt. Featured in this synthesis is an interesting *in situ* hydrogenation reaction utilizing triethyl silane as a source of hydrogen[30]. Slight difficulties with solubility were encountered while removing the Boc group, but this was worked around by simply dissolving the product and washing away any traces of unreacted starting material.



3.9 3.10 3.11

Figure 3.4 Functionalization of linker with alkyne handle.

It was observed that after deprotection, the chloride salt is sensitive and decomposes rapidly, requiring a fast turnaround into the following step. Despite moving from biotin to an alkyne, neoglycosylation to MDP required significant troubleshooting.

Conditions were initially designed based on what was reported in literature by Dr. Bersch, this included a 48-hour mixing period in a pH 4.5 sodium acetate buffer. Initially, this method did not show significant product formation leading to a search for other conditions. Literature displayed fast reaction time with a high yield using a

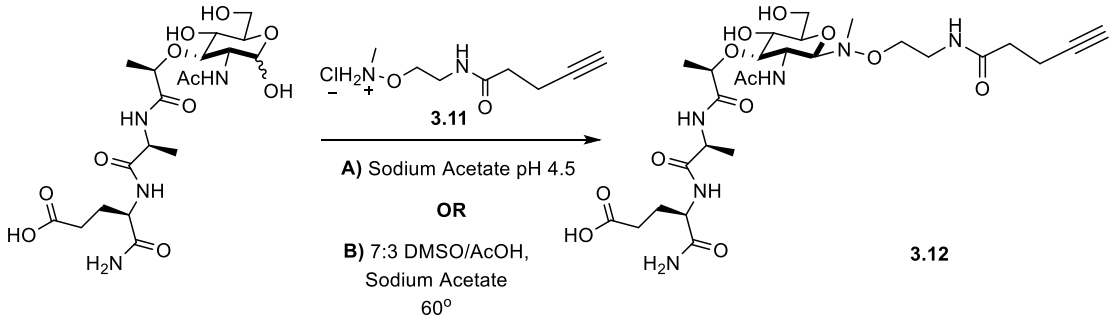


Figure 3.5 Conditions used for attempted neoglycosylation reaction. Condition A is presented in literature by Peri and colleagues[29] as well as by Dr. Bersch[26]. Condition B was also found in literature[31]

7:3 DMSO/AcOH mixture as solvent[31]. This was also attempted but unfortunately, significant product formation was again not observed. In the future, it might be valuable to attempt neoglycosylation to MDP with the alkyne handle removed to prove that the two-carbon linker is compatible with this step.

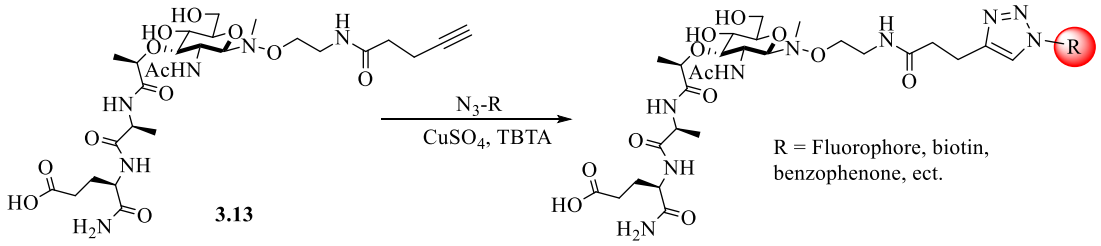


Figure 3.6 Click reaction between MDP-alkyne probe and azide handle of interest

Due to time restrictions, neoglycosylation of the alkyne linker to MDP was not optimized. Based on literature, these reactions simply require further troubleshooting and the exploration of additional conditions. Upon success, copper catalyzed azide alkyne click chemistry (CuAAC) will allow for the rapid installation of any biorthogonal handle of interest.

Chapter 4

DIMER ANALYSIS WITH QRT-PCR

Quantitative reverse transcriptase polymerase chain reaction (qRT-PCR) is a valuable technique for the analysis of gene transcription and activation. Briefly, these assays function through first treating cells with a ligand of interest, in the case of our experiments this ligand would be the dimer compounds we have synthesized. Through collaboration with Professor Hans Christian Reinecker at UT-Southwestern, our compounds will be tested in bone marrow derived macrophages (BMDM's), an excellent model for what might be observed when the gut microbiome interacts with innate immune system. When macrophages are treated with the foreign dimer compounds, cellular machinery alters gene expression, therefore modifying the profile of RNA transcripts in the cell. From here, the enzyme reverse transcriptase is able to convert these RNA's into complimentary DNA (cDNA). With cDNA in hand, sequence specific primers allow for the amplification of designated transcripts. For our purposes, primers will be selected for initial screening based on known inflammatory genes. For example,primers for proteins such as interleukin 6 (IL-6)[32] and CXCL9[33] are good candidates. Then, through application of a fluorescent dye, relative gene expression can be quantified by amplification of cDNA and comparison to a housekeeping gene.

Unfortunately, these experiments were not conducted with my involvement due to time restrictions. It's anticipated that this data will be available in the near future and be featured in the Ph.D thesis of my mentor Siavash Mashayekh.

Chapter 5

CONCLUSIONS AND FUTURE DIRECTIONS

5.1 Synthesis of *m*-DAP Crosslinked Dimers

Here, we have shown a potential synthetic pathway to obtain a 3,3 *m*-DAP linked muramyl dimer, the first of its kind. From here, qRT-PCR will provide valuable insight into the role of DAP dimers in innate immune signaling. It's hypothesized that these compounds will have the ability to activate certain extracellular receptors, such as peptidoglycan binding proteins (PGBP's) and possibly toll like receptors (TLR's). Moreover, this opens the door to additional DAP containing dimers featuring altered modifications or crosslinking patterns. This includes the synthesis of a 3,4 linked DAP dimer as well as a version with $_D$-isoglutamine in the second position as opposed to $_D$-glutamate. Furthermore, other linkages persist in nature such as 2,4. It would be valuable to investigate dimers of this nature. Finally, dimer probes utilizing alkyne and photoaffinity handles would be invaluable for binding and proteomics experiments (See appendix C for schematic).

5.2 Synthesis of a Methyl N,O-Hydroxylamine Linker

Methodology was developed to obtain a short two carbon linker which can be coupled to a biorthogonal handle of interest. This synthesis expanded upon previously published linkers and was aimed at providing fast access to MDP probes. Unfortunately, conditions for neoglycosylation were not optimized due to time constraints. It remains likely that these issues will be resolved in the future with

additional troubleshooting. Once this problem is settled, an interesting application of the linker would be to the dimer compounds we have developed. This would allow for rapid access to probes using the fully deprotected dimer, an aspect which is very valuable in terms of time. Experiments such as photoaffinity labeling or fluorescence tracking would be highly beneficial to further elucidating the role of dimer fragments with the innate immune system (See appendix C for schematic).

Chapter 6

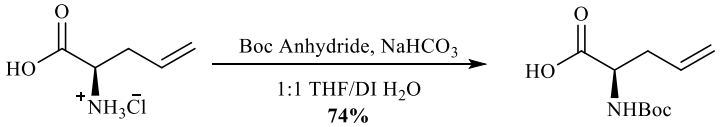
SYNTHETIC METHODOLOGY

6.1 Synthesis of *m*-DAP Dimer Components

Note: Synthesis of 3,3 *m*-DAP dimer was done in heavy collaboration with my graduate mentor. Reaction procedures not noted here can be found in the Ph.D thesis of Siavash Mashayekh (Expected June 2021)

6.1.1 Synthesis of Protected D-Allyl Glycine

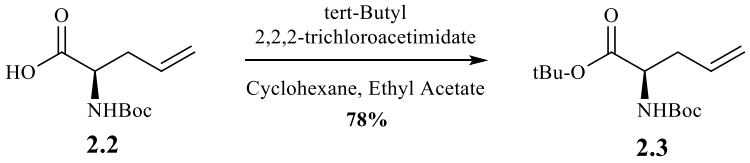
Synthetic scheme is displayed in figure 2.6.



2.1 **2.2**

(R)-2-((*tert*-butoxycarbonyl)amino)pent-4-enoic acid. On ice, D-Allylglycine hydrochloride (0.703 g, 4.65 mmol, 1.0 eq) was dissolved with a 1:1 mixture of DI H_2O and THF (35.8 mL) in a round-bottom flask. Sodium bicarbonate (1.17 g, 14.0 mmol, 3.0 eq) was then added slowly, followed by boc anhydride (1.92 mL, 8.37 mmol, 1.8 eq). The reaction was allowed to rise to room temperature and left to stir for 20 hours. The product was confirmed by LC-MS and then acidified to a pH of 3 with 1 N HCl. The reaction was then extract 4x with ethyl acetate and the organic layer was dried with anhydrous sodium sulfate. After drying under vacuum, the reaction was purified via column chromatography using 0-2% methanol in DCM and 0.1% acetic acid, product presented as a cloudy syrup (734 mg, 74%). ^{1}H NMR (600 MHz,

MeOD) δ 5.81 (ddt, *J* = 17.1, 10.2, 7.0 Hz, 1H), 5.20 – 5.06 (m, 2H), 4.17 (dd, *J* = 8.3, 5.0 Hz, 1H), 2.57 (dt, *J* = 13.0, 6.0 Hz, 1H), 2.48 – 2.39 (m, 1H), 1.46 (s, 9H).

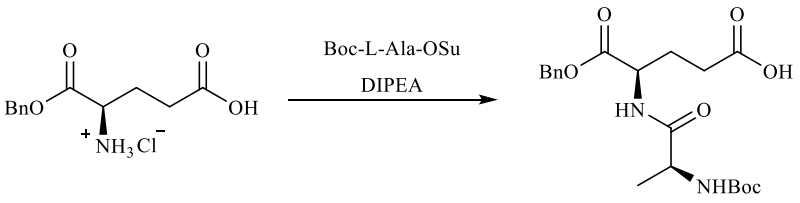


2.2 **2.3**

(R)-*tert*-butyl 2-((*tert*-butoxycarbonyl)amino)pent-4-enoate. In a round-bottom flask, **compound 2.2** (0.708 g, 3.29 mmol, 1 eq) was dissolved in ethyl acetate (15.6 mL). Cyclohexane (7.65 mL) was added followed by *tert*-butyl 2,2,2-trichloroacetimidate (3.6 g, 16.45 mmol, 5 eq) and the reaction was allowed to stir at room temperature for 48 hours. Product was confirmed by LC-MS and the solvent was condensed under vacuum. The reaction was purified via column chromatography using 0-10% ethyl acetate in hexane and 0.1% acetic acid. (Note: product comes off column very quickly.) Product presented as a clear oil (0.697 g, 78%). ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.71 (ddt, *J* = 16.2, 10.5, 7.2 Hz, 1H), 5.15 – 5.09 (m, 2H), 5.04 (d, *J* = 7.7 Hz, 1H), 4.25 (q, *J* = 6.3 Hz, 1H), 2.57 – 2.50 (m, 1H), 2.50 – 2.43 (m, 1H), 1.46 (s, 9H), 1.44 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 171.16, 155.21, 132.58, 118.74, 81.91, 79.62, 53.34, 37.08, 28.34, 28.05.

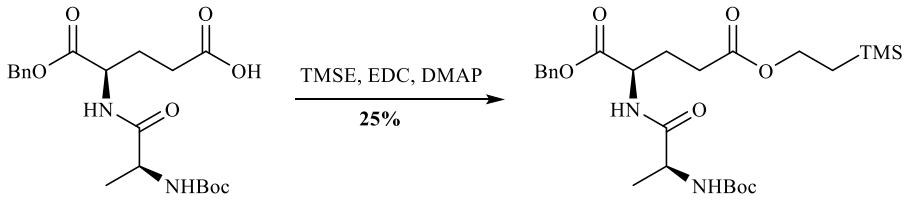
6.1.2 Synthesis of L-Ala-D-Glu Dipeptide

Synthetic scheme shown in figure 2.9.



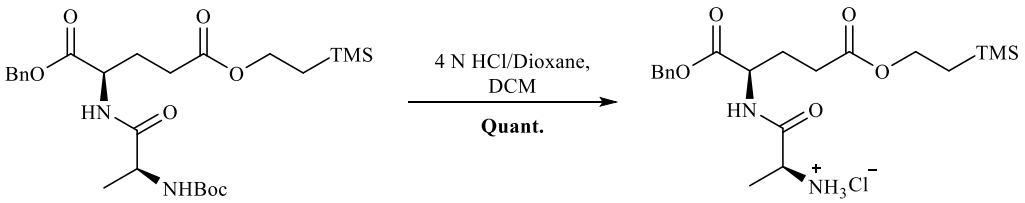
2.15 **2.16**

(R)-5-(benzyloxy)-4-((S)-2-((*tert*-butoxycarbonyl)amino)propanamido)-5-oxopentanoic acid. D-Glutamic acid α-benzyl ester · HCl (500 mg, 1.82 mmol, 1.1 eq) was dissolved in DMF (11 mL) on ice. Boc-L-Ala-OSu (472 mg, 1.65 mmol, 1.0 eq) was added and DIPEA (0.431 mL, 2.48 mmol, 1.5 eq) was dripped in slowly. The reaction was allowed to rise to room temperature and stir overnight. Product was confirmed by LC-MS and TLC (5% MeOH/DCM with 0.1% AcOH and ninhydrin stain). The reaction was diluted with 50 mL ethyl acetate and the organic layer was washed with 25 mL each of 1 N HCl and saturated brine. Solvent was evaporated under reduced pressure and the crude product was purified via silica gel column chromatography (0-3% MeOH/DCM gradient with 0.1% AcOH. Note: purple TLC spot co-elutes with product if gradient is not run very slowly). The product appears as a thin yellow oil (Accurate yield data not collected due to impurity). ^{1}H NMR (600 MHz, MeOD) δ 7.41 – 7.31 (m, 4H), 5.24 – 5.14 (m, 2H), 4.52 (dd, *J* = 9.2, 5.1 Hz, 1H), 4.09 (d, *J* = 8.3 Hz, 1H), 2.38 (t, *J* = 7.5 Hz, 2H), 2.21 (dtd, *J* = 14.1, 7.8, 5.1 Hz, 1H), 1.98 (ddd, *J* = 14.1, 7.0, 2.0 Hz, 1H), 1.46 (s, 9H), 1.30 (d, *J* = 7.2 Hz, 3H).



2.16 2.17

(R)-1-benzyl 5-(2-(trimethylsilyl)ethyl) 2-((S)-2-((*tert*-butoxycarbonyl)amino)propanamido)pentanedioate. On ice, compound **2.16** (881 mg, 2.15 mmol, 1.0 eq) was dissolved in DMF (7.1 mL) and allowed to stir. DMAP (157 mg, 1.29 mmol, 0.6 eq) was added followed by EDC·HCl (452 mg, 2.37 mmol, 1.1 eq) and TMS ethanol (0.380 mL, 2.58 mmol, 1.2 eq) 5 minutes later. The reaction was allowed to rise to room temperature and stir overnight. Product formation was confirmed by LC-MS. The reaction was diluted with 80 mL DCM and then washed with 40 mL each of 1 N HCl, saturated sodium bicarbonate and saturated brine. The organic layer was then dried with sodium sulfate and condensed under reduced pressure. Product was purified via silica gel column chromatography (gradient solvent system of 0-3% MeOH/DCM with 0.1% AcOH) to yield a clear oil (272 mg, 25%). (Note: yield not representative of reaction efficiency, starting material had significant impurity remaining from previous step, largely impacting yield). ^{1}H NMR (600 MHz, MeOD) δ 7.37 – 7.25 (m, 5H), 5.12 (q, *J* = 11.7 Hz, 2H), 4.47 (ddd, *J* = 11.3, 9.1, 5.0 Hz, 1H), 4.16 – 4.09 (m, 2H), 4.03 (s, 1H), 2.33 (dt, *J* = 10.6, 7.1 Hz, 2H), 2.16 (tdd, *J* = 12.9, 9.4, 6.5 Hz, 1H), 1.93 (dddt, *J* = 11.2, 9.1, 7.1, 3.0 Hz, 1H), 1.41 (d, *J* = 10.0 Hz, 9H), 1.25 (dd, *J* = 10.9, 7.1 Hz, 3H), 0.98 – 0.90 (m, 2H), 0.03 – -0.01 (m, 9H).

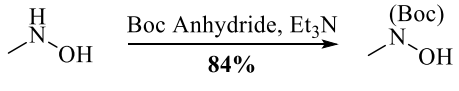


2.17 **2.18**

(R)-1-benzyl 5-(2-(trimethylsilyl)ethyl) 2-((S)-2-aminopropanamido)pentanedioate. (272 mg, 0.535 mmol, 1 eq) was dissolved in DCM (3.5 mL) on ice. 4 N HCl in dioxane (1.33 mL, 5.35 mmol, 10 eq) was then added and the reaction was allowed to stir for one hour. The ice bath was removed and progress was monitored via TLC (5% MeOH/DCM). After four hours, 0.2 mL 4 N HCl in dioxane was added, no starting material was observed an hour later. The reaction was dried under reduced pressure to yield a thick pale-yellow syrup (237 mg, quant.) (Note: Excess dioxane was observed in the NMR spectrum and can be removed through lyophilization). ^{1}H NMR (600 MHz, MeOD) δ 7.35 – 7.24 (m, 5H), 5.14 (s, 2H), 4.50 (dd, *J* = 9.0, 5.4 Hz, 1H), 4.17 – 4.07 (m, 2H), 3.92 (q, *J* = 7.0 Hz, 1H), 2.40 – 2.31 (m, 2H), 2.18 (dtd, *J* = 14.1, 7.7, 5.4 Hz, 1H), 1.99 – 1.90 (m, 1H), 1.47 (d, *J* = 7.1 Hz, 3H), 0.97 – 0.89 (m, 2H), -0.00 (s, 9H).

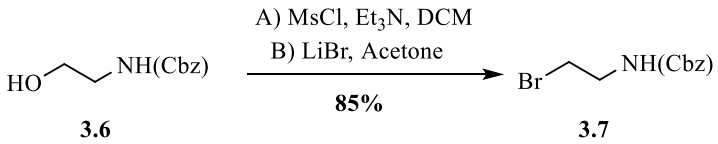
6.2 Synthesis of Methyl N,O-Hydroxylamine Linker

Synthetic schemes shown in figures 3.3 and 3.4.



3.4 **3.5**

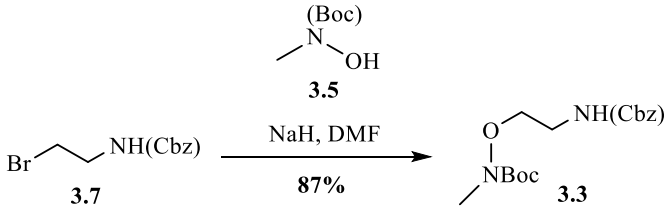
***Tert*-butyl hydroxy(methyl)carbamate.** *N*-Methyl hydroxylamine hydrochloride (9.91 g, 118.65 mmol, 1.3 eq) was dissolved in DCM (296.6 mL). Boc anhydride (21.0 mL, 91.27 mmol, 1 eq) was added followed by triethylamine (12.7 mL, 91.27 mmol, 1.3 eq). (Note: production of gas observed upon addition of triethylamine, ensure flask is properly vented.) Reaction was then allowed to stir overnight. Once complete, the DCM was washed 3x with 750 mL of 0.1 N sodium sulfate. Each aqueous layer was then re-extracted with 200 mL DCM. The organic layer was then condensed under reduced pressure to yield the product as an amber liquid (11.3 g, 84 %). (Note: product evaporates over long periods under reduced pressure, take care while using product on rotary evaporator.). ^{1}H NMR (600 MHz, $CDCl_3$) δ 3.16 (s, 3H), 1.48 (s, 9H).



3.6 **3.7**

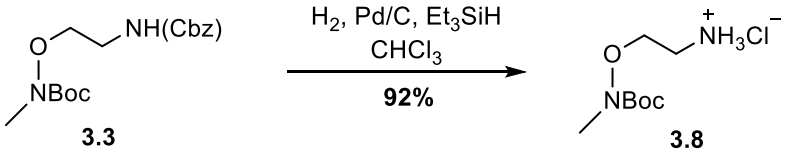
Benzyl (2-bromoethyl)carbamate. On ice, Z-glycinol (5.0g, 25.6 mmol, 1 eq) was dissolved in anhydrous DCM (79 mL). Then, methanesulfonyl chloride (7.37 mL, 30.7 mmol, 1.2 eq) and triethylamine (5.35 mL, 38.4 mmol, 1.5 eq) were added and the reaction was left to stir for one hour on ice. Mesylation was confirmed by TLC (50% ethyl acetate in hexanes with ninhydrin stain). Lithium bromide (26.0 g, 307.2 mmol, 12 eq) was added followed by acetone (51.2 mL) which had been dried on 4 Ångstrom molecular sieves. The reaction was left to stir and rise to room temperature overnight. Bromination was confirmed by TLC (75% ethyl acetate in hexanes). The reaction was

condensed under reduced pressure and dissolved in ethyl acetate. The organic layer was washed with 300 mL each of 1 N HCl, saturated sodium bicarbonate, and saturated brine. The organic layer was dried with sodium sulfate and condensed under reduced pressure. The crude product was purified via silica gel column chromatography (gradient solvent system 0-75% ethyl acetate in hexanes) to yield a yellow oil which crystallizes over long periods left on vacuum. (5.62 g, 85%). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.44 – 7.31 (m, 5H), 5.14 (s, 2H), 3.63 (q, *J* = 5.9 Hz, 2H), 3.49 (t, *J* = 5.9 Hz, 2H).

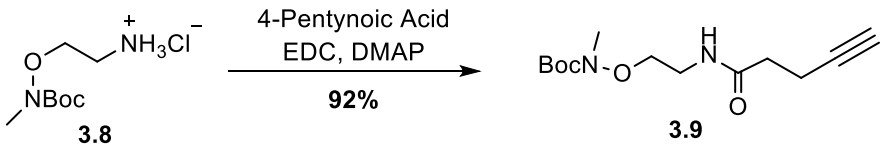


***Tert*-butyl 2-(((benzyloxy)carbonyl)amino)ethoxy(methyl)carbamate.** Compound **3.5** (6.7 mL, 50.3 mmol, 1 eq) was dissolved in anhydrous DMF (100 mL) on ice. 60 % NaH in mineral oil (3.0 g, 75.5 mmol, 1.5 eq) was added slowly and the reaction was left to stir for 5 minutes. Compound **3.7** (sorry Klare idk how to name these) was added (19.5 g, 75.5 mmol, 1.5 eq) and the basicity of the solution was confirmed with litmus paper. Product confirmed via MS after 4 hours and the reaction was quenched with methanol and water. The reaction was condensed and then diluted with 300 mL ethyl acetate. The organic layer was washed with 200 mL each of 1 N HCl, saturated $NaHCO_3$ and brine. The organic layer was condensed under reduced pressure and the crude product was purified via silica gel column chromatography (gradient solvent

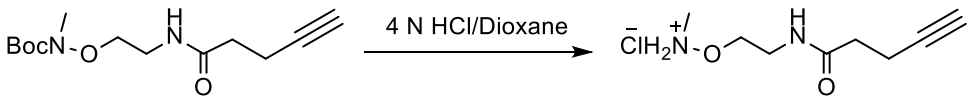
system 0-20% ethyl acetate in hexanes). The pure product presents as a light brown syrup (14.23 g, 87%). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.42 – 7.28 (m, 5H), 5.13 (s, 2H), 3.90 (t, *J* = 4.8 Hz, 2H), 3.43 (q, *J* = 5.3 Hz, 2H), 3.09 (s, 3H), 1.50 (s, 9H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 157.42, 156.53, 136.65, 128.47, 128.05, 82.01, 72.99, 66.63, 39.39, 36.65, 28.23.



(((*tert*-butoxycarbonyl)(methyl)amino)oxy)ethanaminium chloride. Compound **3.8** (680 mg, 2.09 mmol, 1eq) was dissolved in anhydrous methanol (14 mL) and allowed to stir. Pd/C (10% by mass) (222 mg, 0.209 mmol, 0.1 eq) was added carefully, followed by triethyl silane (3.3 mL, 20.9 mmol, 10 eq). Note: reaction vessel must be fit with both an N_2 line and vent needle to prevent pressure buildup. Chloroform (0.200 mL, 2.51 mmol, 1.2 eq) was added and the reaction was allowed to stir for one hour. Progress was monitored by TLC (5% MeOH/DCM with ninhydrin stain) until no starting material remained. The reaction was vacuum filtered over celite and solvent was removed under reduced pressure to yield a flakey white solid (0.471 g, 92 %). ^{1}H NMR (600 MHz, MeOD) δ 4.10-4.09 (m, 2H), 3.19-3.17 (m, 2H), 3.16 (s, 3H), 1.53 (s, 9H).



***Tert*-butyl methyl(2-(pent-4-ynamido)ethoxy)carbamate**. On ice, 4-pentynoic acid (18.1 mg, 0.185 mmol, 1 eq) was dissolved in DMF (0.61 mL). DMAP was added, followed by EDC and the reaction was allowed to stir for 5 minutes. Then, compound **3.9** (50 mg, 0.222 mmol, 1.2 eq) was added in portions over another 5 minutes. The reaction was allowed to rise to room temperature and stir overnight. Formation of product was confirmed by LC-MS and TLC (5% MeOH/DCM with ninhydrin stain). The reaction was diluted with 5 mL ethyl acetate and the organic layer was washed with 2.5 mL each of 1N HCl, saturated sodium bicarbonate, and saturated brine. Product was detected via TLC in the aqueous layers and they were each reextracted with 10 mL DCM to provide the full yield. Product presented as a pale yellow oil (46 mg, 92%). ^{1}H NMR (600 MHz, MeOD) δ 3.87 (t, *J* = 5.4 Hz, 2H), 3.38 (t, *J* = 5.4 Hz, 2H), 3.07 (s, 3H), 2.48 – 2.42 (m, 2H), 2.39 (td, *J* = 7.0, 1.4 Hz, 2H), 2.23 (t, *J* = 2.6 Hz, 1H), 1.46 (s, 9H). ^{13}C NMR (151 MHz, MeOD) δ 172.70, 157.41, 82.08, 81.61, 72.31, 68.88, 37.77, 37.66, 35.70, 34.63, 27.12, 14.24.

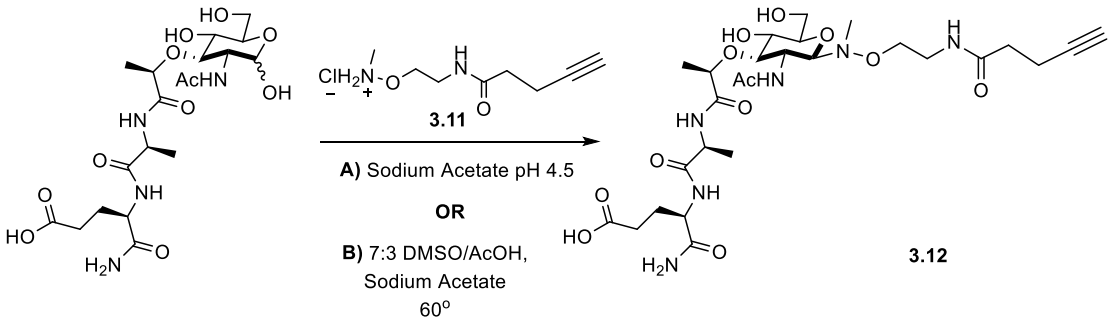


3.9 **3.10**

N-methyl-O-(2-(pent-4-ynamido)ethyl)hydroxylammonium chloride. Compound 3.9 (24 mg, 0.088 mmol, 1 eq) was dissolved in DCM (0.2 mL) on ice in a 10 mL RBF. 4 N HCl in dioxane (0.45 mL) was then added and the reaction was stirred on ice for 30 minutes. The ice bath was removed and the reaction was allowed to stir for

an additional 3 hours, progress was monitored by TLC (5% MeOH/DCM with ninhydrin stain). During the course of the reaction, product precipitated as a sticky grey solid. Once starting material was consumed, the reaction was diluted with 4 mL 1 N HCl and washed with 6 mL DCM twice. The aqueous layer was lyophilized to yield a clear yellow oil (18 mg, quantitative). Note: product is unstable and should be used shortly after drying. Decomposition was noted if the appearance of the product changed to a sticky brown solid. Excess acid and dioxane can be removed via further lyophilization. ^{1}H NMR (600 MHz, D_2O) δ 4.04 (t, *J* = 5.2 Hz, 2H), 3.39 (t, *J* = 5.2 Hz, 2H), 2.86 (s, 3H), 2.39 – 2.30 (m, 4H), 2.23 (s, 1H).

6.3 Neoglycosylation Between MDP and Alkyne Linker



MDP-Alkyne. A) In a 10 mL round bottom flask, compound **3.11** (45 mg, 0.218 mmol, 10 eq.) was dissolved in 1.0 M pH 4.5 sodium acetate buffer (0.22 mL). MDP (10.7 mg, 0.0218 mmol, 1 eq.) was added and the reaction was allowed to stir for ~48 hours. LC-MS showed the presence of product but significant starting material remained. (Note: Purification not completed).

B) Compound **3.11** (48 mg, 0.233 mmol, 10 eq) was dissolved with DMSO (0.63 mL0 and acetic acid (0.27 mL). The mixture was then transferred to a 10 mL round bottom flask which contained MDP (11.4 mg, 0.0233 mmol, 1 eq.). Sodium acetate (19.1 mg, 0.233 mmol, 10 eq.) was added and the reaction was heated at 60 ^{o}C for two hours. Temperature was then lowered to 40 ^{o}C and the reaction was allowed to stir for ~48 hours. LC-MS showed the presence of product but significant starting material remained. (Purification not completed).

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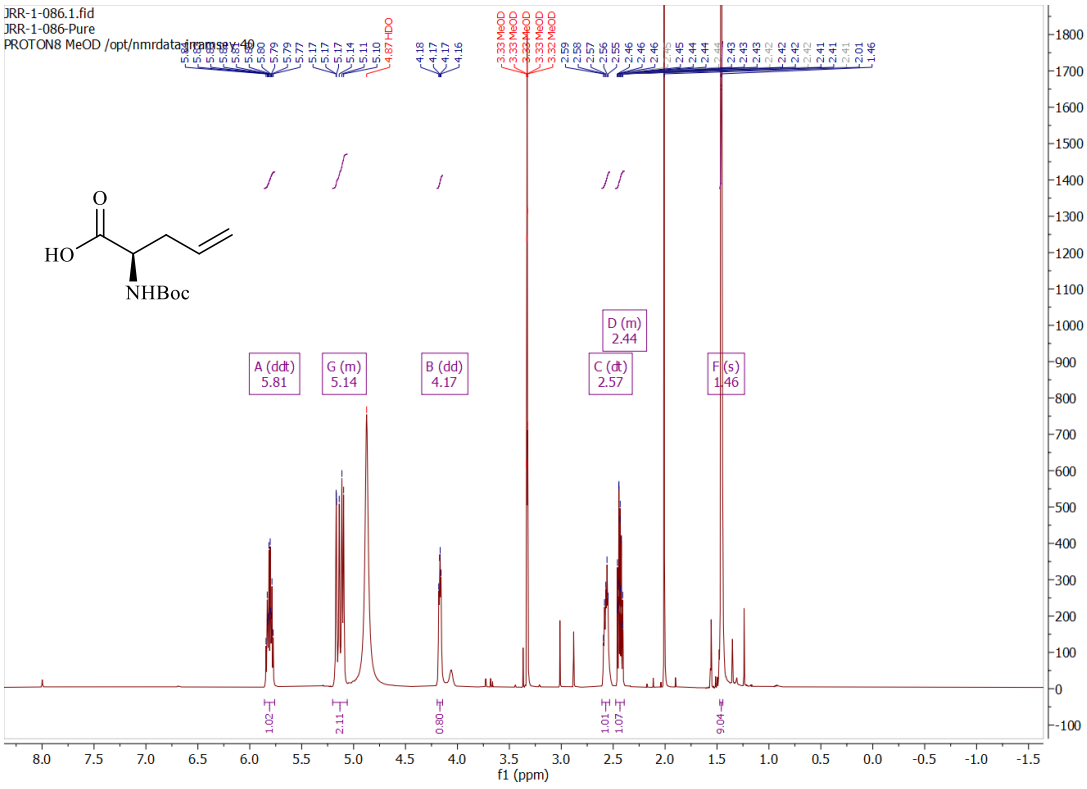
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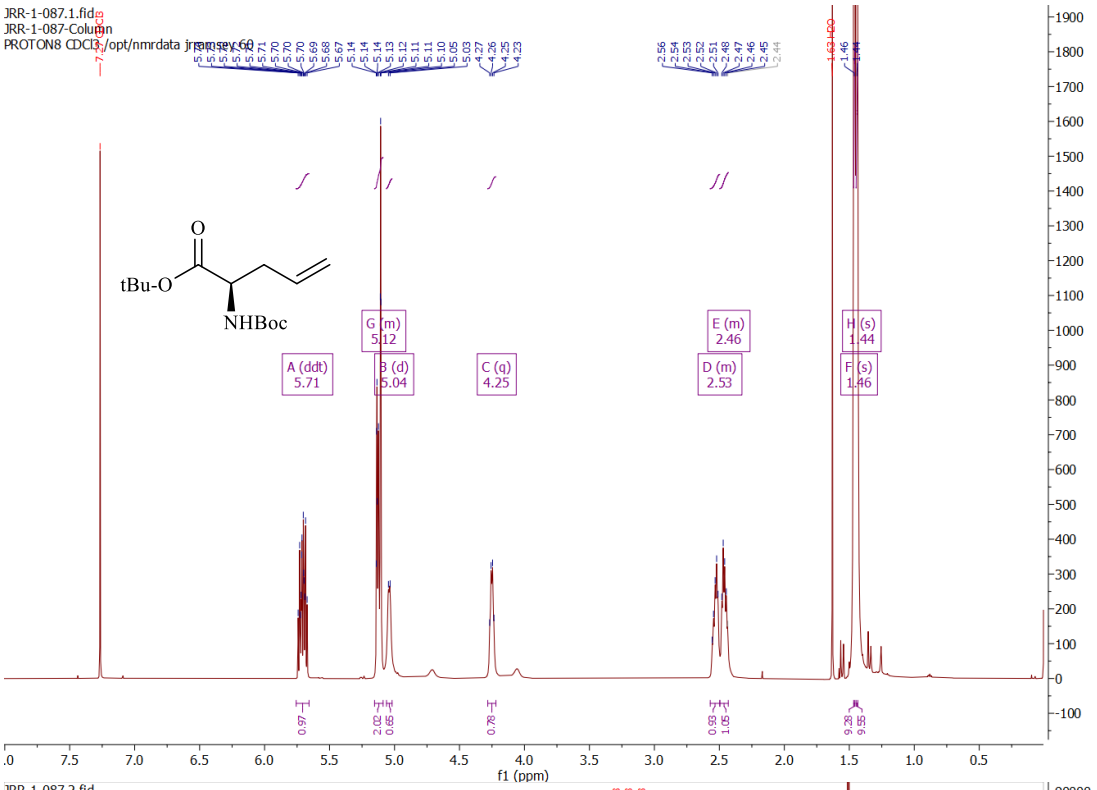
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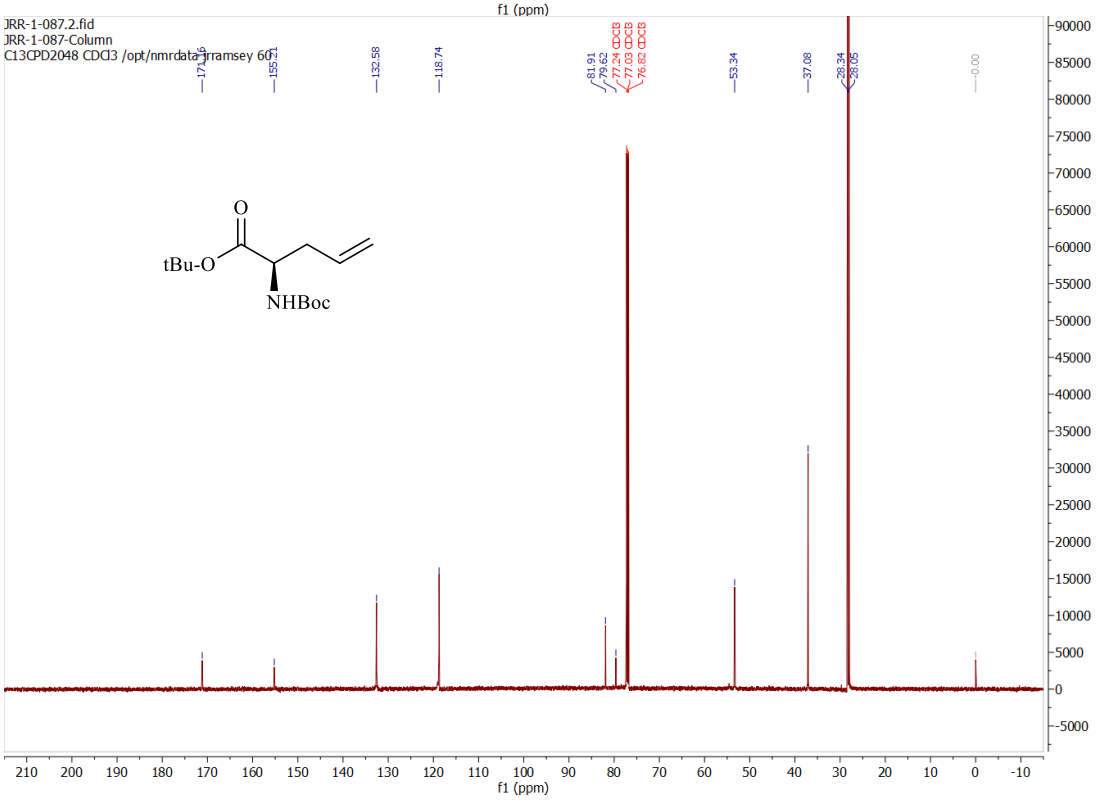
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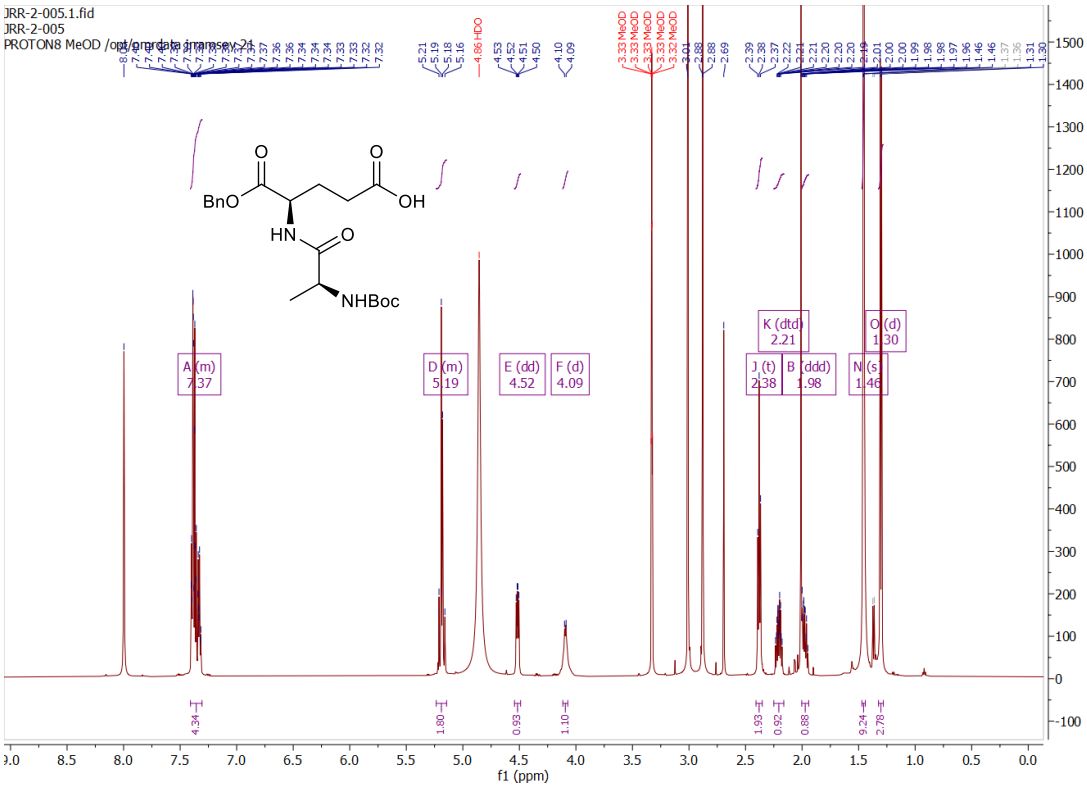
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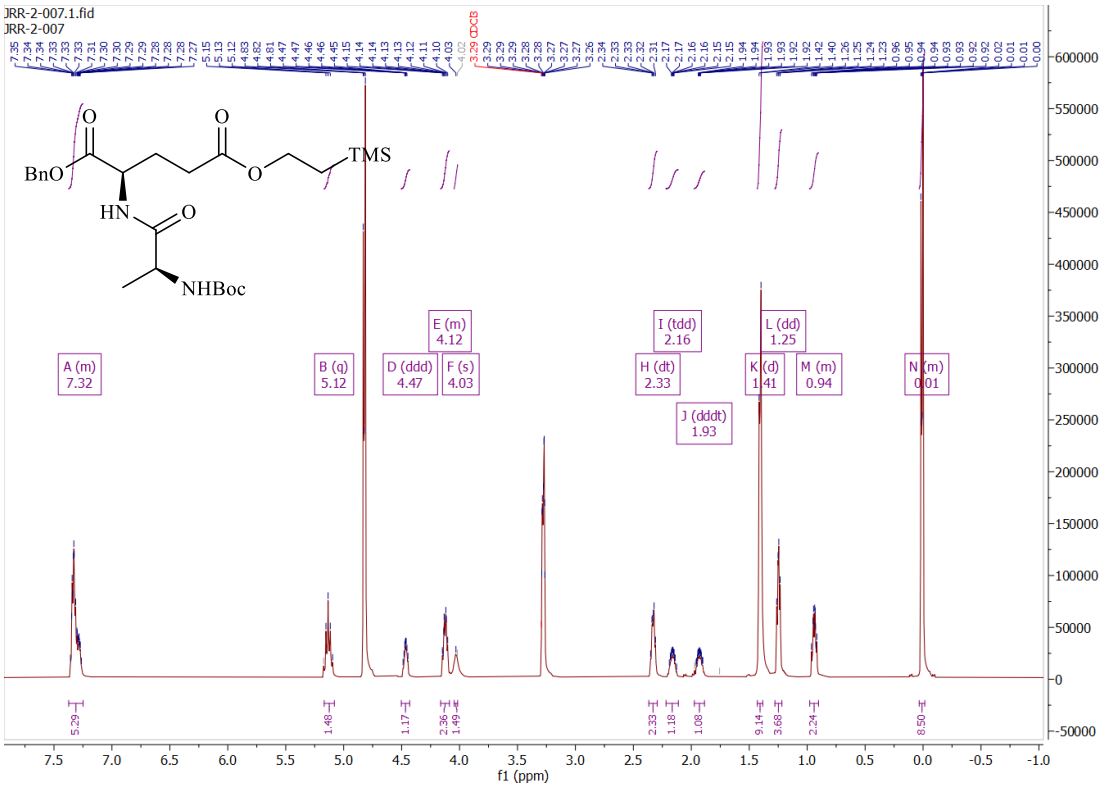
NMR SPECTRA

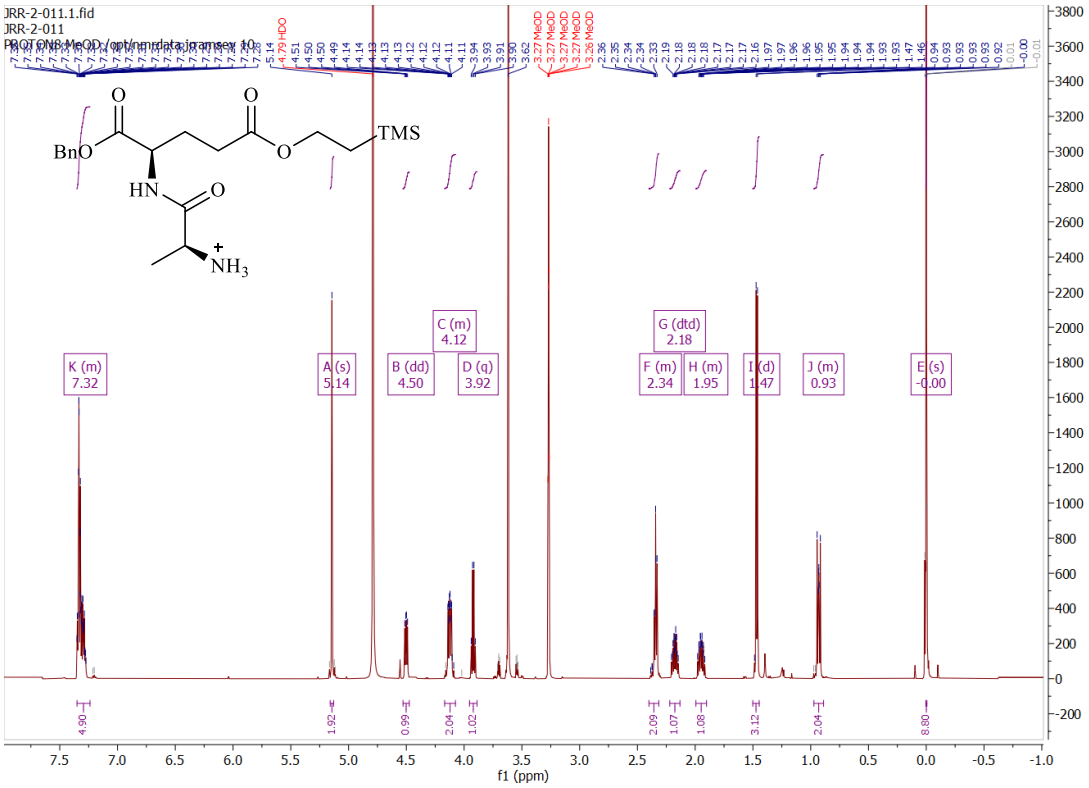


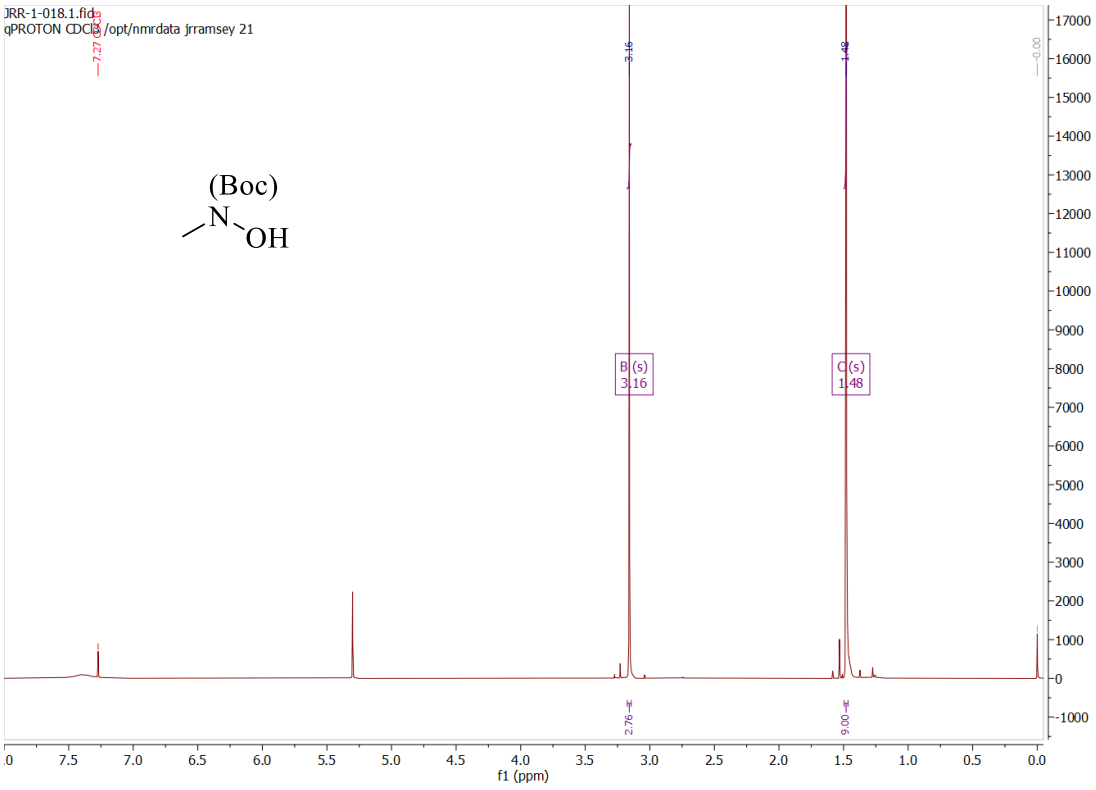

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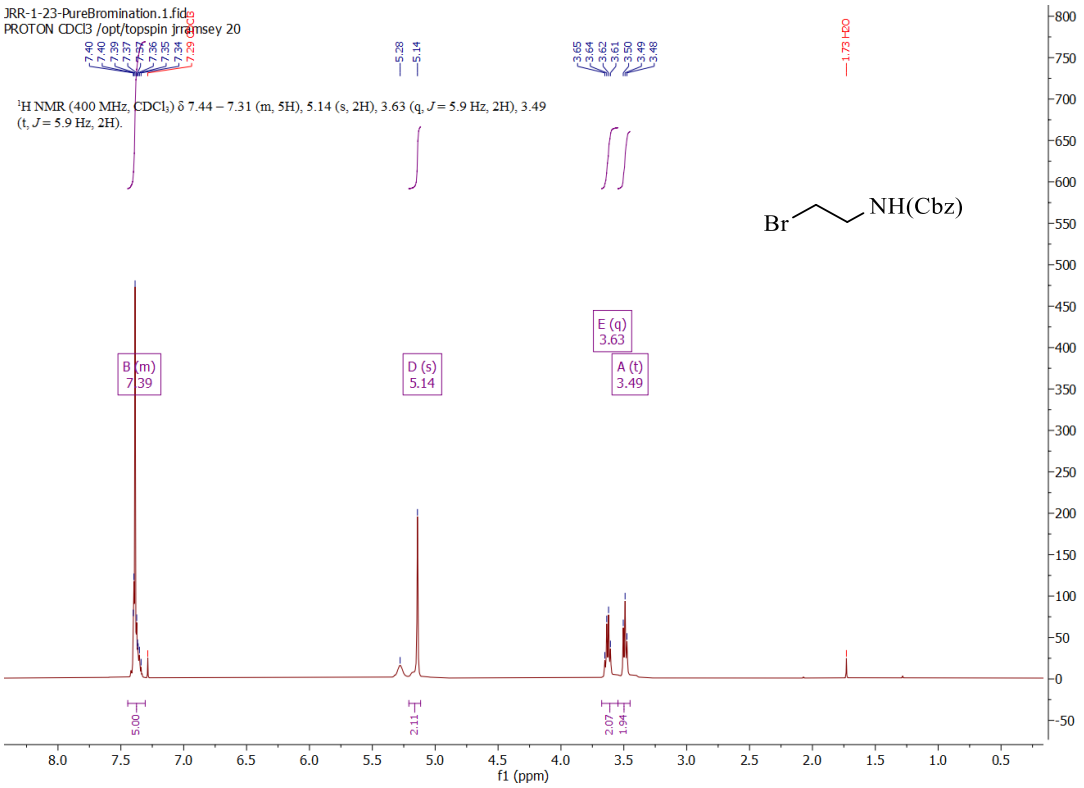

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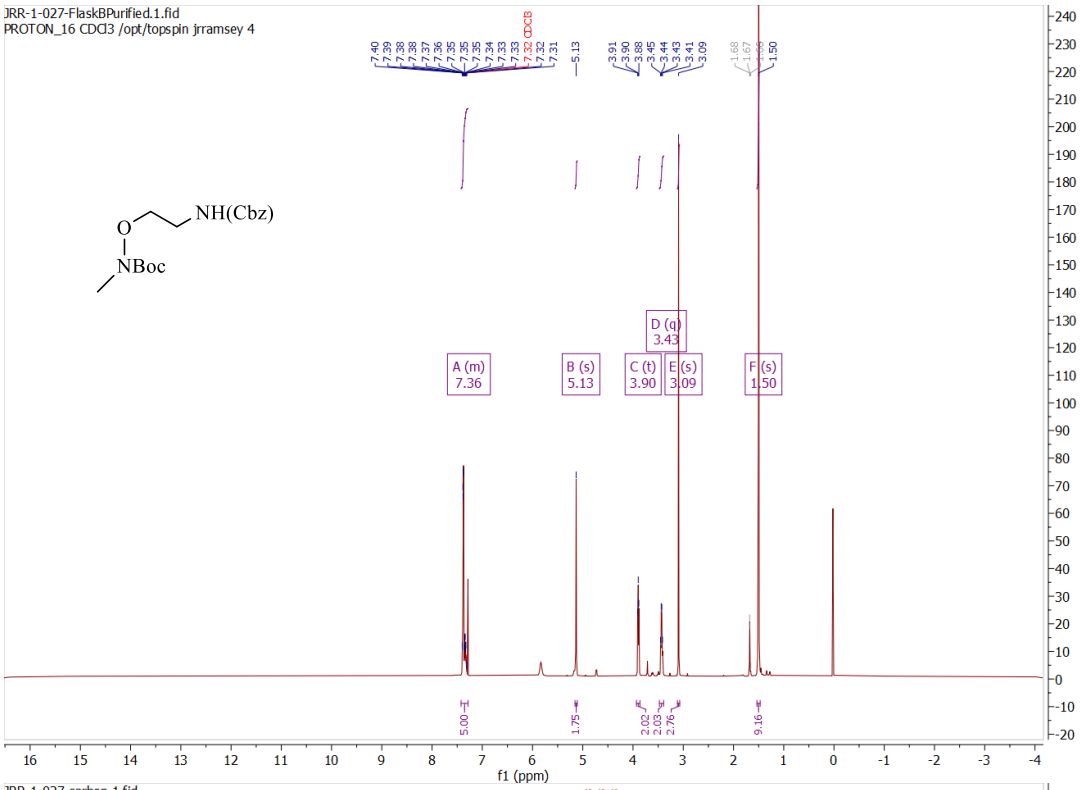

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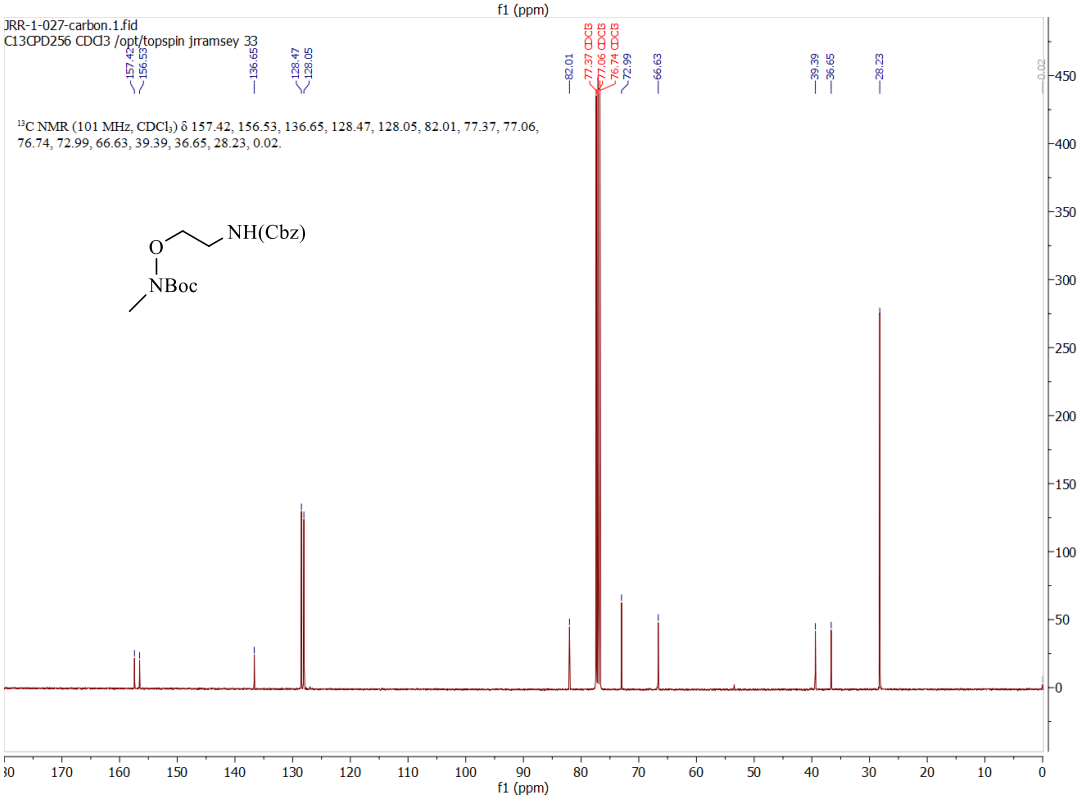

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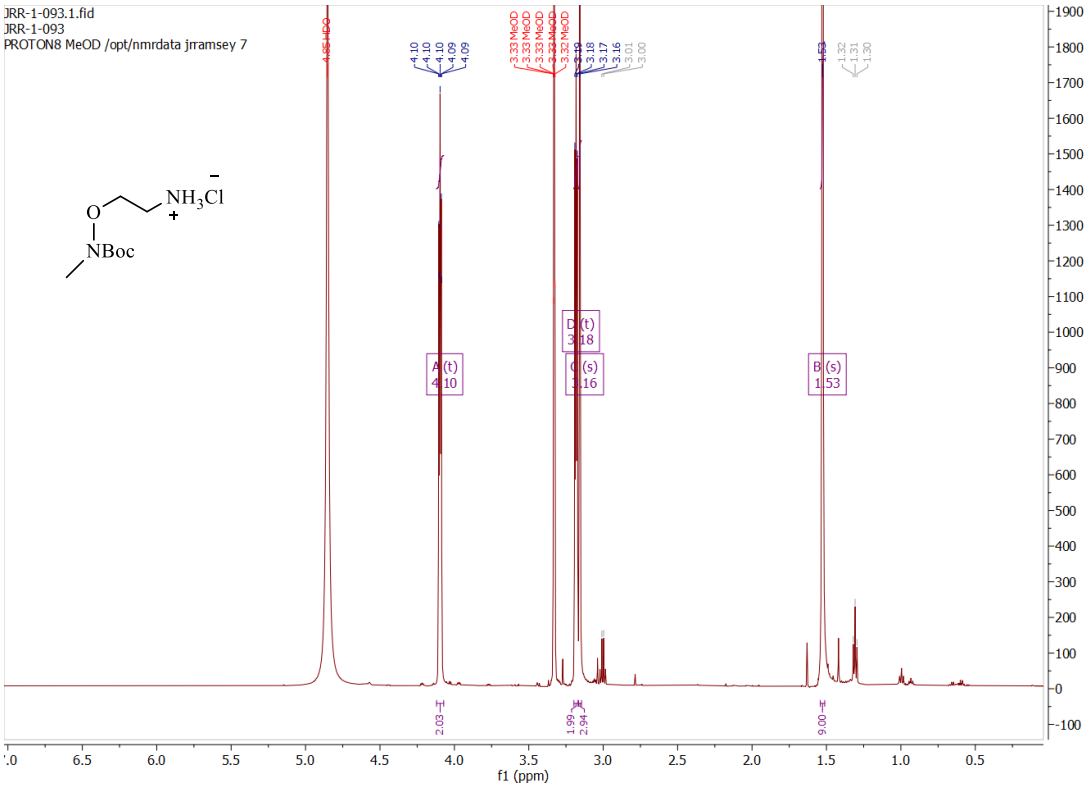

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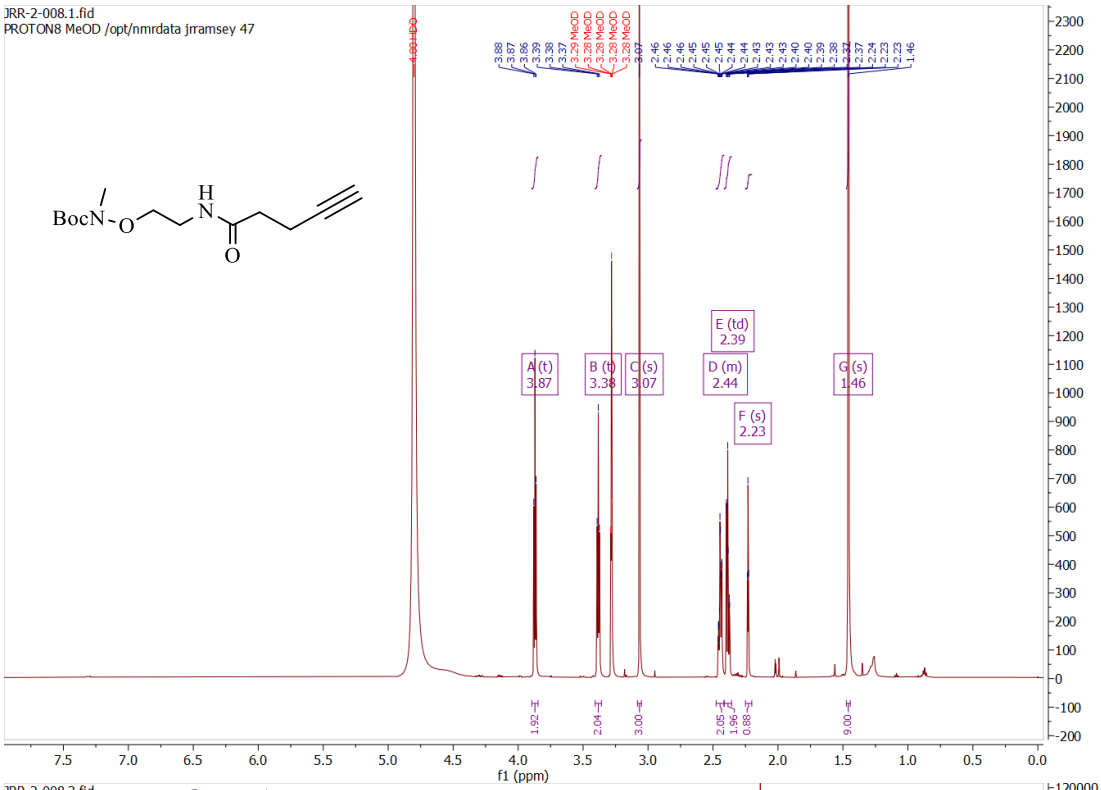

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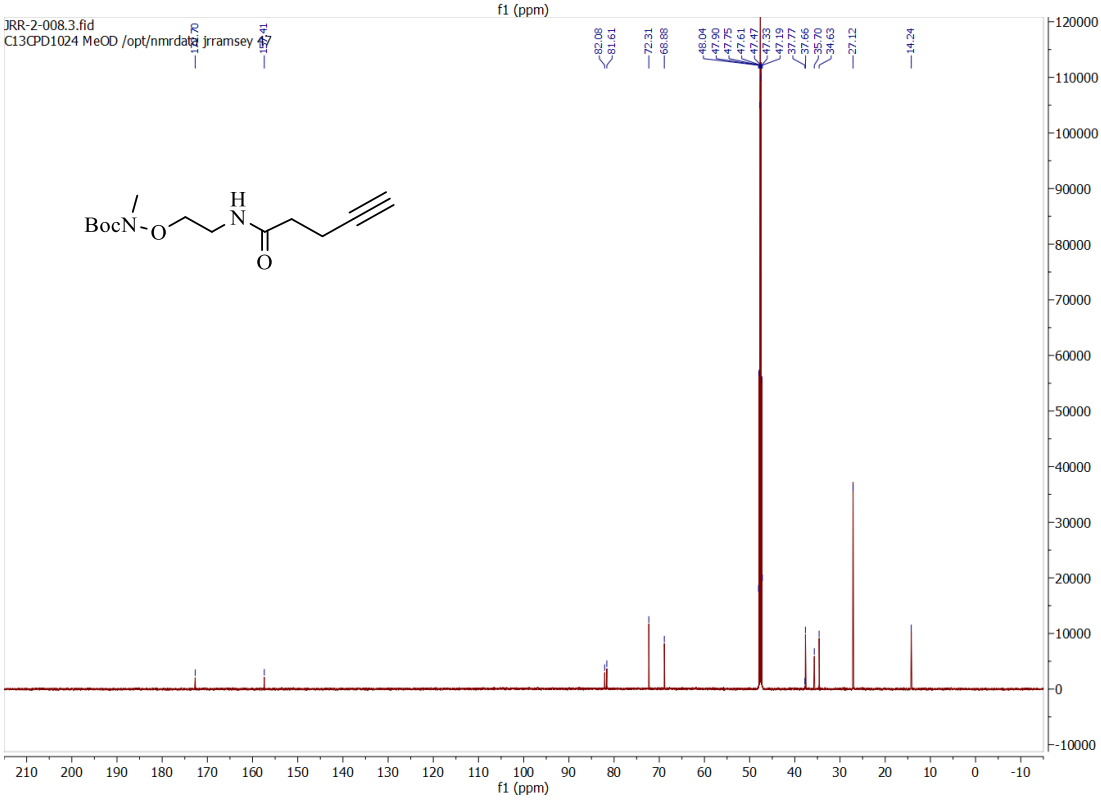

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NH(Cbz)
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D (s) 5.14
E (q) 3.63
A (t) 3.49
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1.94
f1 (ppm)

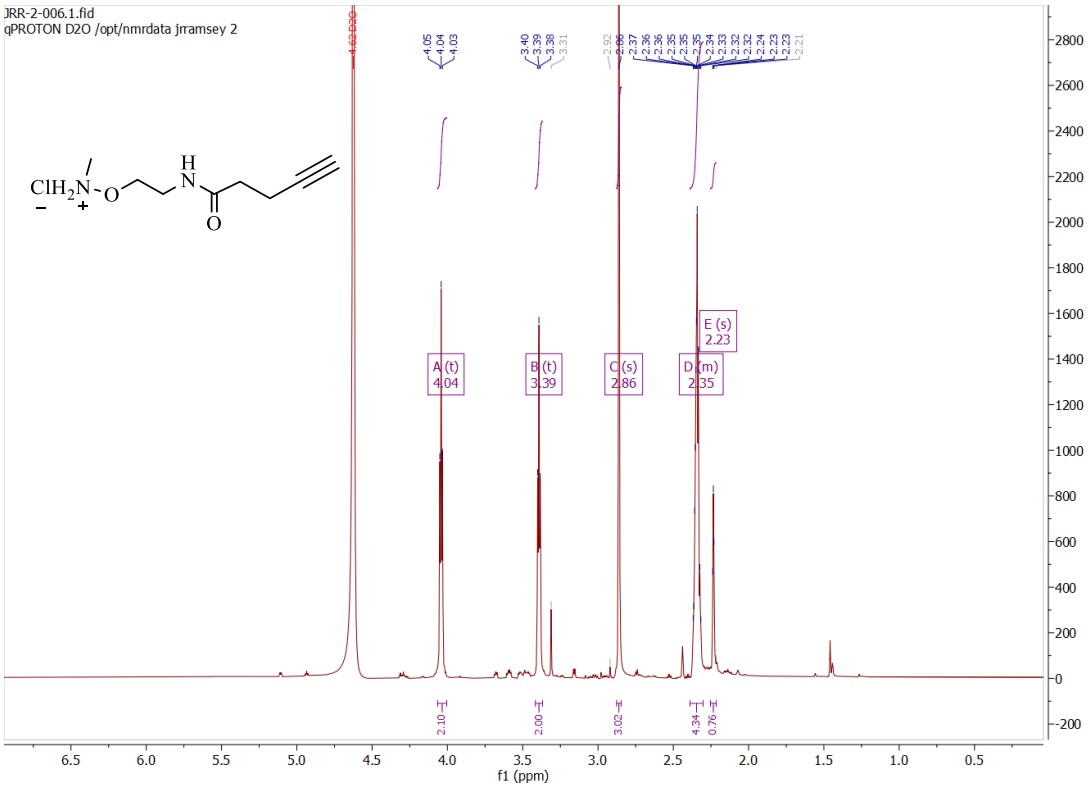

JRR-1-027-FlaskBPurified.1.fid
PROTON_16 CDCl3 /opt/topspin jrramsey 4
NH(Cbz)
O
NBoc
A (m) 7.36
B (s) 5.13
C (t) 3.90
D (q) 3.43
E (s) 3.09
F (s) 1.50
f1 (ppm)


JRR-1-027-carbon.1.fid
C13CPD256 CDCl3 /opt/topspin jrramsey 33
^{13}C NMR (101 MHz, $CDCl_3$) δ 157.42, 156.53, 136.65, 128.47, 128.05, 82.01, 77.37, 77.06, 76.74, 72.99, 66.63, 39.39, 36.65, 28.23, 0.02.
NH(Cbz)
O
NBoc
f1 (ppm)


JRR-1-093.1.fid
JRR-1-093
PROTON8 MeOD /opt/nmrdata jrramsey 7
NH3Cl
O
NBoc
A (t) 4.10
B (s) 1.53
C (s) 3.16
D (t) 3.18
f1 (ppm)


JRR-2-008.1.fid
PROTON8 MeOD /opt/nmrdata jrramsey 47
A (t) 3.87
B (t) 3.38
C (s) 3.07
D (m) 2.44
E (td) 2.39
F (s) 2.23
G (s) 1.46
f1 (ppm)

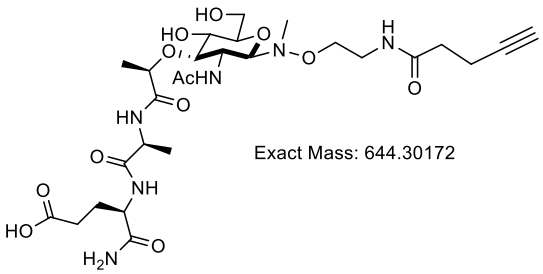

JRR-2-008.3.fid
C13CPD1024 MeOD /opt/nmrdata jrramsey 47
f1 (ppm)


JRR-2-006.1.fid
qPROTON D2O /opt/nmrdata jrramsey 2
A (t)
4.04
B (t)
3.39
C (s)
2.86
D (m)
2.35
E (s)
2.23
2.10
2.00
3.02
4.34
0.76
f1 (ppm)

Appendix B

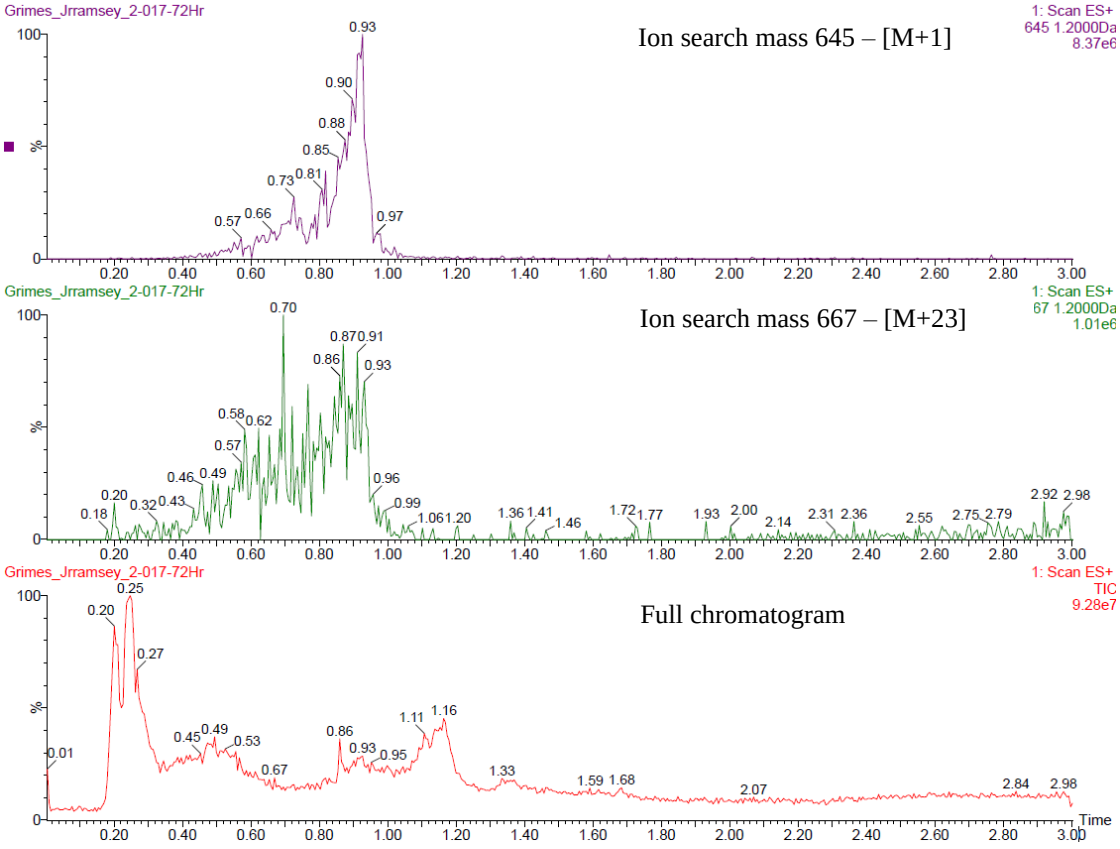
MASS SPECTRA

LC-MS Chromatograms from reactions shown in figure 3.5

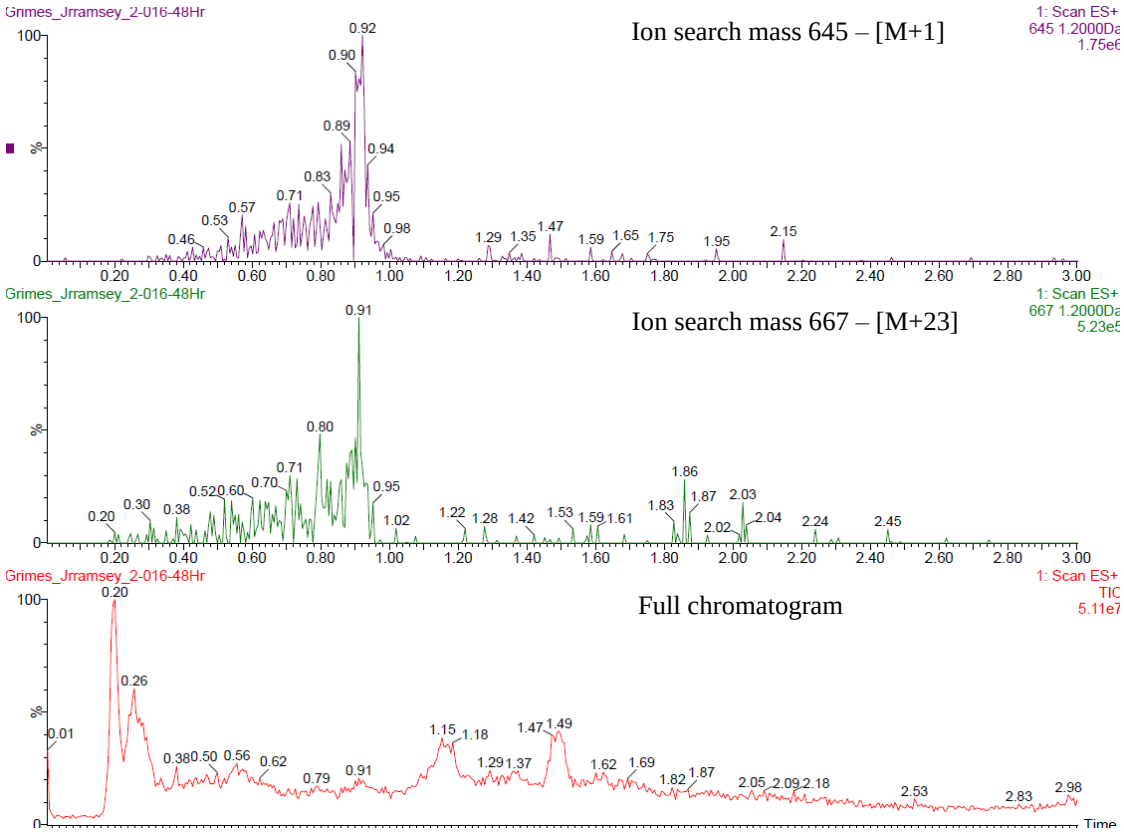


Exact Mass: 644.30172

Conditions **A)**



Conditions **B)**



Appendix C

LIBRARY OF POTENTIAL DIMERS AND PROBES

R_1 = OH, or NH_2. Green indicates compounds which have been investigated in the past, red indicates possible directions for future synthesis[8].

